

Link.Foundation.Links.Notation Class Library

1.1 ./Link.Foundation.Links.Notation/FormatConfig.cs

```
1 namespace Link.Foundation.Links.Notation
2 {
3     /// <summary>
4     /// FormatConfig is an alias for FormatOptions to maintain API consistency
5     /// with the Python implementation.
6     ///
7     /// C# uses 'FormatOptions' internally, but provides 'FormatConfig'
8     /// as an alias to match Python's naming convention.
9     /// </summary>
10    public class FormatConfig : FormatOptions
11    {
12        // FormatConfig inherits all functionality from FormatOptions
13        // No additional implementation needed - this is purely for naming consistency
14    }
15 }
```

1.2 ./Link.Foundation.Links.Notation/FormatOptions.cs

```
1 using System;
2
3 namespace Link.Foundation.Links.Notation
4 {
5     /// <summary>
6     /// FormatOptions for Lino notation formatting.
7     /// Provides configuration options for controlling how Link objects are formatted.
8     /// </summary>
9     public class FormatOptions
10    {
11        /// <summary>
12        /// If true, omit parentheses where safe (default: false)
13        /// </summary>
14        public bool LessParentheses { get; set; } = false;
15
16        /// <summary>
17        /// Maximum line length before auto-indenting (default: 80)
18        /// </summary>
19        public int MaxLineLength { get; set; } = 80;
20
21        /// <summary>
22        /// If true, indent lines exceeding MaxLineLength (default: false)
23        /// </summary>
24        public bool IndentLongLines { get; set; } = false;
25
26        /// <summary>
27        /// Maximum number of references before auto-indenting (default: null = unlimited)
28        /// </summary>
29        public int? MaxInlineRefs { get; set; } = null;
30
31        /// <summary>
32        /// If true, group consecutive links with same ID (default: false)
33        /// </summary>
34        public bool GroupConsecutive { get; set; } = false;
35
36        /// <summary>
37        /// String to use for indentation (default: "  " = two spaces)
38        /// </summary>
39        public string IndentString { get; set; } = "  ";
40
41        /// <summary>
42        /// If true, prefer inline format when under thresholds (default: true)
43        /// </summary>
44        public bool PreferInline { get; set; } = true;
45
46        /// <summary>
47        /// Check if line should be indented based on length
48        /// </summary>
49        /// <param name="line">The line to check</param>
50        /// <returns>True if line should be indented based on length threshold</returns>
51        public bool ShouldIndentByLength(string line)
52        {
53            if (!IndentLongLines)
54            {
55                return false;
56            }
57            // Count printable unicode characters
58            return line.Length > MaxLineLength;
59        }
60    }
```

```

61     /// <summary>
62     /// Check if link should be indented based on reference count
63     /// </summary>
64     /// <param name="refCount">Number of references in the link</param>
65     /// <returns>True if link should be indented based on reference count threshold</returns>
66     public bool ShouldIndentByRefCount(int refCount)
67     {
68         if (MaxInlineRefs == null)
69         {
70             return false;
71         }
72         return refCount > MaxInlineRefs.Value;
73     }
74 }
75 }

```

1.3 ./Link.Foundation.Links.Notation/ILinksGroupListExtensions.cs

```

1  using System.Collections.Generic;
2  using System.Runtime.CompilerServices;
3
4  namespace Link.Foundation.Links.Notation
5  {
6      /// <summary>
7      /// Provides extension methods for collections of <see cref="LinksGroup{TLinkAddress}"/>
8      instances.
9      /// </summary>
10     public static class ILinksGroupListExtensions
11     {
12         /// <summary>
13         /// Converts a collection of links groups to a flat list of links by processing each
14         group hierarchically.
15         /// </summary>
16         /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This
17         can be any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
18         /// <param name="groups">The collection of links groups to convert.</param>
19         /// <returns>A flat list of links representing all the links from the input
20         groups.</returns>
21         [MethodImpl(MethodImplOptions.AggressiveInlining)]
22         public static List<Link<TLinkAddress>> ToLinksList<TLinkAddress>(this
23         IList<LinksGroup<TLinkAddress>> groups)
24         {
25             var list = new List<Link<TLinkAddress>>();
26             for (var i = 0; i < groups.Count; i++)
27             {
28                 CollectLinksWithIndentedIdSyntaxSupport(list, groups[i], null);
29             }
30             return list;
31         }
32
33         private static void
34         CollectLinksWithIndentedIdSyntaxSupport<TLinkAddress>(List<Link<TLinkAddress>> list,
35         LinksGroup<TLinkAddress> group, Link<TLinkAddress>? parentDependency)
36         {
37             var link = group.Link;
38             var groups = group.Groups;
39
40             if (groups != null && groups.Count > 0)
41             {
42                 bool isIndentedIdSyntax = link.Id != null &&
43                     (link.Values == null || link.Values.Count == 0);
44
45                 if (isIndentedIdSyntax && !parentDependency.HasValue)
46                 {
47                     var childValues = new List<Link<TLinkAddress>>();
48                     for (int i = 0; i < groups.Count; i++)
49                     {
50                         var transformedLink = TransformIndentedIdLink(groups[i]);
51                         childValues.Add(transformedLink);
52                     }
53
54                     var linkWithChildren = new Link<TLinkAddress>(link.Id, childValues);
55                     list.Add(linkWithChildren);
56                 }
57                 else
58                 {
59                     var transformedGroups = new List<LinksGroup<TLinkAddress>>();
60                     for (int i = 0; i < groups.Count; i++)
61                     {
62                         var childGroup = groups[i];

```

```

56         var childLink = childGroup.Link;
57         var childGroups = childGroup.Groups;
58
59         if (childLink.Id != null &&
60             (childLink.Values == null || childLink.Values.Count == 0) &&
61             childGroups != null && childGroups.Count > 0)
62         {
63             var transformedLink = TransformIndentedIdLink(childGroup);
64             transformedGroups.Add(new LinksGroup<TLinkAddress>(transformedLink));
65         }
66         else
67         {
68             transformedGroups.Add(childGroup);
69         }
70     }
71
72     var currentDependency = parentDependency.HasValue ?
↪ parentDependency.Value.Combine(link) : link;
73     list.Add(currentDependency);
74
75     for (int i = 0; i < transformedGroups.Count; i++)
76     {
77         CollectLinksWithIndentedIdSyntaxSupport(list, transformedGroups[i],
↪ currentDependency);
78     }
79 }
80 }
81 else
82 {
83     var currentLink = parentDependency.HasValue ?
↪ parentDependency.Value.Combine(link) : link;
84     list.Add(currentLink);
85 }
86 }
87
88 private static Link<TLinkAddress>
↪ TransformIndentedIdLink<TLinkAddress>(LinksGroup<TLinkAddress> group)
89 {
90     var link = group.Link;
91     var groups = group.Groups;
92
93     if (groups != null && groups.Count > 0 &&
94         link.Id != null &&
95         (link.Values == null || link.Values.Count == 0))
96     {
97         var childValues = new List<Link<TLinkAddress>>();
98         for (int i = 0; i < groups.Count; i++)
99         {
100             var transformedChild = TransformIndentedIdLink(groups[i]);
101             childValues.Add(transformedChild);
102         }
103         return new Link<TLinkAddress>(link.Id, childValues);
104     }
105     else if (!group.IsParenthesized && link.Values != null && link.Values.Count == 1)
106     {
107         return link.Values[0];
108     }
109     else
110     {
111         return link;
112     }
113 }
114 }
115 }

```

1.4 ./Link.Foundation.Links.Notation/IListExtensions.cs

```

1 using Platform.Collections;
2 using System;
3 using System.Collections.Generic;
4 using System.Linq;
5 using System.Runtime.CompilerServices;
6
7 namespace Link.Foundation.Links.Notation
8 {
9     /// <summary>
10     /// Provides extension methods for formatting collections of <see
↪ cref="Link{TLinkAddress}" /> instances.
11     /// </summary>
12     public static class IListExtensions

```

```

13 {
14     /// <summary>
15     /// Formats a collection of links as a multi-line string with each link on a separate
16     line.
17     /// </summary>
18     /// <typeparam name="TLinkAddress">The type used for link
19     addresses/identifiers.</typeparam>
20     /// <param name="links">The collection of links to format.</param>
21     /// <returns>A multi-line string representation of the links.</returns>
22     [MethodImpl(MethodImplOptions.AggressiveInlining)]
23     public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links) =>
24     string.Join(Environment.NewLine, links.Select(l => l.ToString()));
25
26     /// <summary>
27     /// Formats a collection of links as a multi-line string with optional parentheses
28     trimming for cleaner output.
29     /// </summary>
30     /// <typeparam name="TLinkAddress">The type used for link
31     addresses/identifiers.</typeparam>
32     /// <param name="links">The collection of links to format.</param>
33     /// <param name="lessParentheses">True to remove outer parentheses from each link for
34     cleaner formatting; false to use standard formatting.</param>
35     /// <returns>A multi-line string representation of the links.</returns>
36     [MethodImpl(MethodImplOptions.AggressiveInlining)]
37     public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links, bool
38     lessParentheses)
39     {
40         if (lessParentheses == false)
41         {
42             return links.Format();
43         }
44         else
45         {
46             return string.Join(Environment.NewLine, links.Select(l =>
47             l.ToString().TrimSingle('(').TrimSingle(')')));
48         }
49     }
50
51     /// <summary>
52     /// Formats a collection of links using FormatOptions configuration.
53     /// </summary>
54     /// <typeparam name="TLinkAddress">The type used for link
55     addresses/identifiers.</typeparam>
56     /// <param name="links">The collection of links to format.</param>
57     /// <param name="options">The FormatOptions to use for formatting.</param>
58     /// <returns>A multi-line string representation of the links.</returns>
59     [MethodImpl(MethodImplOptions.AggressiveInlining)]
60     public static string Format<TLinkAddress>(this IList<Link<TLinkAddress>> links,
61     FormatOptions options)
62     {
63         if (links == null || links.Count == 0)
64         {
65             return string.Empty;
66         }
67
68         // Apply consecutive link grouping if enabled
69         var linksToFormat = options.GroupConsecutive ? GroupConsecutiveLinks(links) : links;
70
71         // Format each link with options
72         var formattedLinks = linksToFormat.Select(l => l.FormatWithOptions(options));
73         return string.Join(Environment.NewLine, formattedLinks);
74     }
75
76     /// <summary>
77     /// Group consecutive links with the same ID.
78     /// </summary>
79     private static List<Link<TLinkAddress>>
80     GroupConsecutiveLinks<TLinkAddress>(IList<Link<TLinkAddress>> links)
81     {
82         if (links == null || links.Count == 0)
83         {
84             return new List<Link<TLinkAddress>>();
85         }
86
87         var grouped = new List<Link<TLinkAddress>>();
88         int i = 0;
89
90         while (i < links.Count)

```

```

80     {
81         var current = links[i];
82
83         // Look ahead for consecutive links with same ID
84         if (current.Id != null && current.Values != null && current.Values.Count > 0)
85         {
86             // Collect all values with same ID
87             var sameIdValues = new List<Link<TLinkAddress>>(current.Values);
88             int j = i + 1;
89
90             while (j < links.Count)
91             {
92                 var nextLink = links[j];
93                 if (EqualityComparer<TLinkAddress>.Default.Equals(nextLink.Id,
↪ current.Id) &&
94                     nextLink.Values != null && nextLink.Values.Count > 0)
95                 {
96                     sameIdValues.AddRange(nextLink.Values);
97                     j++;
98                 }
99                 else
100                 {
101                     break;
102                 }
103             }
104
105             // If we found consecutive links, create grouped link
106             if (j > i + 1)
107             {
108                 var groupedLink = new Link<TLinkAddress>(current.Id, sameIdValues);
109                 grouped.Add(groupedLink);
110                 i = j;
111                 continue;
112             }
113         }
114
115         grouped.Add(current);
116         i++;
117     }
118
119     return grouped;
120 }
121 }
122 }

```

1.5 ./Link.Foundation.Links.Notation/Link.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Runtime.CompilerServices;
4  using System.Text;
5  using Platform.Collections;
6  using Platform.Collections.Lists;
7
8  namespace Link.Foundation.Links.Notation
9  {
10     /// <summary>
11     /// Represents a link in the Links Notation with an optional identifier and nested values.
12     /// Links can represent simple references, complex nested structures, or relationships
↪ between references to links.
13     /// </summary>
14     /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
↪ any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
15     public struct Link<TLinkAddress> : IEquatable<Link<TLinkAddress>>
16     {
17         private static readonly EqualityComparer<TLinkAddress> EqualityComparerInstance =
↪ EqualityComparer<TLinkAddress>.Default;
18
19         /// <summary>
20         /// Gets the identifier or address of this link. Can be null for anonymous links.
21         /// </summary>
22         public readonly TLinkAddress? Id;
23
24         /// <summary>
25         /// Gets the collection of nested link values. Can be null or empty for simple reference
↪ links.
26         /// </summary>
27         public readonly IList<Link<TLinkAddress>>? Values;
28
29         /// <summary>
30         /// Initializes a new link with the specified identifier and values.

```

```

31     /// </summary>
32     /// <param name="id">The link identifier or address.</param>
33     /// <param name="values">The nested link values.</param>
34     [MethodImpl(MethodImplOptions.AggressiveInlining)]
35     public Link(TLinkAddress? id, IList<Link<TLinkAddress>>? values) => (Id, Values) = (id,
↪ values);

36
37     /// <summary>
38     /// Initializes a new anonymous link with the specified values.
39     /// </summary>
40     /// <param name="values">The nested link values.</param>
41     [MethodImpl(MethodImplOptions.AggressiveInlining)]
42     public Link(IList<Link<TLinkAddress>> values) : this(default!, values) { }

43
44     /// <summary>
45     /// Initializes a new anonymous link with the specified array of values.
46     /// </summary>
47     /// <param name="values">The nested link values as a parameter array.</param>
48     [MethodImpl(MethodImplOptions.AggressiveInlining)]
49     public Link(params Link<TLinkAddress>[] values) : this(default!, values) { }

50
51     /// <summary>
52     /// Initializes a new simple reference link with the specified identifier.
53     /// </summary>
54     /// <param name="id">The link identifier or address.</param>
55     [MethodImpl(MethodImplOptions.AggressiveInlining)]
56     public Link(TLinkAddress id) : this(id, default!) { }

57
58     /// <summary>
59     /// Returns the string representation of this link in Links Notation format.
60     /// </summary>
61     /// <returns>A string representation of the link with proper escaping and
↪ formatting.</returns>
62     [MethodImpl(MethodImplOptions.AggressiveInlining)]
63     public override string ToString() => Id == null ?
64         $"({GetValuesString()})" :
65         (Values == null || Values.Count == 0) ?
66             $"({EscapeReference(Id.ToString())})" :
67             $"({EscapeReference(Id.ToString())}: {GetValuesString()})";

68
69     /// <summary>
70     /// Gets the string representation of the link's values.
71     /// </summary>
72     /// <returns>A space-separated string of the values, or empty string if no values
↪ exist.</returns>
73     [MethodImpl(MethodImplOptions.AggressiveInlining)]
74     public string GetValuesString()
75     {
76         if (Values == null || Values.Count == 0)
77         {
78             return "";
79         }
80         var sb = new StringBuilder();
81         for (int i = 0; i < Values.Count; i++)
82         {
83             if (i > 0)
84             {
85                 sb.Append(' ');
86             }
87             sb.Append(GetValueString(Values[i]));
88         }
89         return sb.ToString();
90     }

91
92     /// <summary>
93     /// Simplifies the link structure by unwrapping single-value containers and recursively
↪ simplifying nested values.
94     /// </summary>
95     /// <returns>A simplified version of this link.</returns>
96     [MethodImpl(MethodImplOptions.AggressiveInlining)]
97     public Link<TLinkAddress> Simplify()
98     {
99         if (Values == null || Values.Count == 0)
100         {
101             return this;
102         }
103         else if (Values.Count == 1)
104         {

```

```

105         return Values[0];
106     }
107     else
108     {
109         var newValues = new Link<TLinkAddress>[Values.Count];
110         for (int i = 0; i < Values.Count; i++)
111         {
112             newValues[i] = Values[i].Simplify();
113         }
114         return new Link<TLinkAddress>(Id, newValues);
115     }
116 }
117
118 /// <summary>
119 /// Combines this link with another link to create a new link containing both as values.
120 /// </summary>
121 /// <param name="other">The other link to combine with.</param>
122 /// <returns>A new link containing both links as values.</returns>
123 [MethodImpl(MethodImplOptions.AggressiveInlining)]
124 public Link<TLinkAddress> Combine(Link<TLinkAddress> other) => new
↪ Link<TLinkAddress>(this, other);
125
126 /// <summary>
127 /// Gets the string representation of a link value, choosing the appropriate format.
128 /// </summary>
129 /// <param name="value">The link value to convert to string.</param>
130 /// <returns>The string representation of the link value.</returns>
131 [MethodImpl(MethodImplOptions.AggressiveInlining)]
132 public static string GetValueString(Link<TLinkAddress> value) =>
↪ value.ToLinkOrIdString();
133
134 /// <summary>
135 /// Escapes a reference string for safe use in Links Notation format by adding quotes if
↪ necessary.
136 /// </summary>
137 /// <param name="reference">The reference string to escape.</param>
138 /// <returns>The escaped reference string with appropriate quoting.</returns>
139 public static string EscapeReference(string? reference)
140 {
141     if (reference == null)
142     {
143         return "";
144     }
145     // The empty reference is written as a bare delimiter pair, so that it reads
146     // back as itself instead of disappearing from the document.
147     if (reference.Length == 0)
148     {
149         return "\"\"";
150     }
151     if (
152         reference.Contains(":") ||
153         reference.Contains("(") ||
154         reference.Contains(")") ||
155         reference.Contains(" ") ||
156         reference.Contains("\t") ||
157         reference.Contains("\n") ||
158         reference.Contains("\r") ||
159         reference.Contains("\"")
160     )
161     {
162         return $"{reference}'";
163     }
164     else if (reference.Contains("'"))
165     {
166         return $"{reference}\"";
167     }
168     else
169     {
170         return reference;
171     }
172 }
173
174 /// <summary>
175 /// Converts the link to its string representation, choosing between simple reference
↪ format or full link format.
176 /// </summary>
177 /// <returns>A string representation optimized for the link's content.</returns>
178 [MethodImpl(MethodImplOptions.AggressiveInlining)]

```

```

179         public string ToLinkOrIdString() => (Values == null || Values.Count == 0) ? (Id == null
↳ ? "" : EscapeReference(Id?.ToString())) : ToString();
180
181         /// <summary>
182         /// Implicitly converts a value to a simple reference link.
183         /// </summary>
184         /// <param name="value">The value to convert.</param>
185         [MethodImpl(MethodImplOptions.AggressiveInlining)]
186         public static implicit operator Link<TLinkAddress>(TLinkAddress value) => new
↳ Link<TLinkAddress>(value);
187
188         /// <summary>
189         /// Implicitly converts a tuple of identifier and values to a link.
190         /// </summary>
191         /// <param name="value">The tuple containing identifier and values.</param>
192         [MethodImpl(MethodImplOptions.AggressiveInlining)]
193         public static implicit operator Link<TLinkAddress>((TLinkAddress,
↳ IList<Link<TLinkAddress>>) value) => new(value.Item1, value.Item2);
194
195         /// <summary>
196         /// Implicitly converts a source-target pair to an anonymous link.
197         /// </summary>
198         /// <param name="value">The tuple containing source and target links.</param>
199         [MethodImpl(MethodImplOptions.AggressiveInlining)]
200         public static implicit operator Link<TLinkAddress>((Link<TLinkAddress> source,
↳ Link<TLinkAddress> target) value) => new(value.source, value.target);
201
202         /// <summary>
203         /// Implicitly converts a tuple of identifier, source, and target to a link.
204         /// </summary>
205         /// <param name="value">The tuple containing id, source and target.</param>
206         [MethodImpl(MethodImplOptions.AggressiveInlining)]
207         public static implicit operator Link<TLinkAddress>((id<TLinkAddress> id,
↳ Link<TLinkAddress> source, Link<TLinkAddress> target) value) => new(value.id.Id, new[] {
↳ value.source, value.target });
208
209         /// <summary>
210         /// Implicitly converts a source-linker-target triplet to an anonymous link.
211         /// </summary>
212         /// <param name="value">The tuple containing source, linker, and target links.</param>
213         [MethodImpl(MethodImplOptions.AggressiveInlining)]
214         public static implicit operator Link<TLinkAddress>((Link<TLinkAddress> source,
↳ Link<TLinkAddress> linker, Link<TLinkAddress> target) value) => new(value.source,
↳ value.linker, value.target);
215
216         /// <summary>
217         /// Implicitly converts a tuple of identifier, source, linker, and target to a link.
218         /// </summary>
219         /// <param name="value">The tuple containing id, source, linker, and target.</param>
220         [MethodImpl(MethodImplOptions.AggressiveInlining)]
221         public static implicit operator Link<TLinkAddress>((id<TLinkAddress> id,
↳ Link<TLinkAddress> source, Link<TLinkAddress> linker, Link<TLinkAddress> target) value) =>
↳ new(value.id.Id, new[] { value.source, value.linker, value.target });
222
223         /// <summary>
224         /// Determines whether the specified object is equal to this link.
225         /// </summary>
226         /// <param name="obj">The object to compare with this link.</param>
227         /// <returns>True if the specified object is equal to this link; otherwise,
↳ false.</returns>
228         [MethodImpl(MethodImplOptions.AggressiveInlining)]
229         public override bool Equals(object? obj) => obj is Link<TLinkAddress> link &&
↳ Equals(link);
230
231         /// <summary>
232         /// Returns the hash code for this link.
233         /// </summary>
234         /// <returns>A hash code for this link.</returns>
235         [MethodImpl(MethodImplOptions.AggressiveInlining)]
236         public override int GetHashCode() => (Id, (Values ??
↳ Array.Empty<Link<TLinkAddress>>()).GenerateHashCode()).GetHashCode();
237
238         /// <summary>
239         /// Indicates whether the current link is equal to another link.
240         /// Two anonymous links (both with null Ids) are considered equal if their values are
↳ equal.
241         /// </summary>

```



```

242     /// <param name="other">The link to compare with this link.</param>
243     /// <returns>True if the current link is equal to the other parameter; otherwise,
↪ false.</returns>
244     [MethodImpl(MethodImplOptions.AggressiveInlining)]
245     public bool Equals(Link<TLinkAddress> other)
246     {
247         // Both have null IDs - compare only values
248         if (Id == null && other.Id == null)
249         {
250             return (Values ?? Array.Empty<Link<TLinkAddress>>()).EqualTo(other.Values ??
↪ Array.Empty<Link<TLinkAddress>>());
251         }
252
253         // Only one has null ID - not equal
254         if (Id == null || other.Id == null)
255         {
256             return false;
257         }
258
259         // Both have IDs - compare IDs and values
260         return EqualityComparerInstance.Equals(Id, other.Id) &&
261             (Values ?? Array.Empty<Link<TLinkAddress>>()).EqualTo(other.Values ??
↪ Array.Empty<Link<TLinkAddress>>());
262     }
263
264     /// <summary>
265     /// Determines whether two link instances are equal.
266     /// </summary>
267     /// <param name="left">The first link to compare.</param>
268     /// <param name="right">The second link to compare.</param>
269     /// <returns>True if the links are equal; otherwise, false.</returns>
270     [MethodImpl(MethodImplOptions.AggressiveInlining)]
271     public static bool operator ==(Link<TLinkAddress> left, Link<TLinkAddress> right) =>
↪ left.Equals(right);
272
273     /// <summary>
274     /// Determines whether two link instances are not equal.
275     /// </summary>
276     /// <param name="left">The first link to compare.</param>
277     /// <param name="right">The second link to compare.</param>
278     /// <returns>True if the links are not equal; otherwise, false.</returns>
279     [MethodImpl(MethodImplOptions.AggressiveInlining)]
280     public static bool operator !=(Link<TLinkAddress> left, Link<TLinkAddress> right) =>
↪ !(left == right);
281 }
282 }

```

1.6 ./Link.Foundation.Links.Notation/LinkFormatExtensions.cs

```

1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Runtime.CompilerServices;
5  using System.Text;
6
7  namespace Link.Foundation.Links.Notation
8  {
9      /// <summary>
10     /// Provides extension methods for formatting <see cref="Link{TLinkAddress}"/> instances
↪ with FormatOptions.
11     /// </summary>
12     public static class LinkFormatExtensions
13     {
14         /// <summary>
15         /// Format a link using FormatOptions configuration.
16         /// </summary>
17         /// <typeparam name="TLinkAddress">The type used for link
↪ addresses/identifiers.</typeparam>
18         /// <param name="link">The link to format.</param>
19         /// <param name="options">The FormatOptions to use.</param>
20         /// <returns>Formatted string representation.</returns>
21         public static string FormatWithOptions<TLinkAddress>(this Link<TLinkAddress> link,
↪ FormatOptions options)
22         {
23             // Empty link
24             if (link.Id == null && (link.Values == null || link.Values.Count == 0))
25             {
26                 return options.LessParentheses ? "" : "()";
27             }

```

```

28
29 // Link with only ID, no values
30 if (link.Values == null || link.Values.Count == 0)
31 {
32     var escapedId = Link<TLinkAddress>.EscapeReference(link.Id?.ToString());
33     return options.LessParentheses && !NeedsParentheses(link.Id?.ToString()) ?
34         escapedId :
35         $"({escapedId})";
36 }
37
38 // Check if we should use indented format
39 bool shouldIndent = false;
40 if (options.ShouldIndentByRefCount(link.Values.Count))
41 {
42     shouldIndent = true;
43 }
44 else
45 {
46     // Try inline format first
47     var valuesStr = string.Join(" ", link.Values.Select(v => GetValueString(v)));
48     string testLine;
49     if (link.Id != null)
50     {
51         var idStr = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
52         testLine = options.LessParentheses ? $"({idStr}: {valuesStr})" : $"({idStr}:
↪ {valuesStr})";
53     }
54     else
55     {
56         testLine = options.LessParentheses ? valuesStr : $"({valuesStr})";
57     }
58
59     if (options.ShouldIndentByLength(testLine))
60     {
61         shouldIndent = true;
62     }
63 }
64
65 // Format with indentation if needed
66 if (shouldIndent && options.PreferInline == false)
67 {
68     return FormatIndented(link, options);
69 }
70
71 // Standard inline formatting
72 var values = string.Join(" ", link.Values.Select(v => GetValueString(v)));
73
74 // Link with values only (null id)
75 if (link.Id == null)
76 {
77     if (options.LessParentheses)
78     {
79         var allSimple = link.Values.All(v => v.Values == null || v.Values.Count ==
↪ 0);
80         if (allSimple)
81         {
82             return string.Join(" ", link.Values.Select(v =>
↪ Link<TLinkAddress>.EscapeReference(v.Id?.ToString())));
83         }
84         return values;
85     }
86     return $"({values})";
87 }
88
89 // Link with ID and values
90 var id = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
91 var withColon = $"({id}: {values})";
92 return options.LessParentheses && !NeedsParentheses(link.Id?.ToString()) ?
93     withColon :
94     $"({withColon})";
95 }
96
97 /// <summary>
98 /// Format a link with indentation.
99 /// </summary>
100 private static string FormatIndented<TLinkAddress>(Link<TLinkAddress> link,
↪ FormatOptions options)
101 {
102     // FormatIndented is only reached with a non-empty Values collection,

```

```

103 // but the nullable annotation on Link<T>.Values does not know that.
104 var values = link.Values ?? Array.Empty<Link<TLinkAddress>>();
105
106 if (link.Id == null)
107 {
108     // Values only - format each on separate line
109     var lines = values.Select(v => options.IndentString + GetValueString(v));
110     return string.Join(Environment.NewLine, lines);
111 }
112
113 // Link with ID - format as id:\n value1\n value2
114 var idStr = Link<TLinkAddress>.EscapeReference(link.Id.ToString());
115 var sb = new StringBuilder();
116 sb.Append($"{idStr}:");
117
118 foreach (var v in values)
119 {
120     sb.Append(Environment.NewLine);
121     sb.Append(options.IndentString);
122     sb.Append(GetValueString(v));
123 }
124
125 return sb.ToString();
126 }
127
128 /// <summary>
129 /// Get the string representation of a value.
130 /// </summary>
131 private static string GetValueString<TLinkAddress>(Link<TLinkAddress> value)
132 {
133     return value.ToLinkOrIdString();
134 }
135
136 /// <summary>
137 /// Check if a string needs to be wrapped in parentheses.
138 /// </summary>
139 private static bool NeedsParentheses(string? s)
140 {
141     return s != null && (s.Contains(" ") || s.Contains(":") || s.Contains("(") ||
142     ↪ s.Contains(")"));
143 }
144 }

```

1.7 ./Link.Foundation.Links.Notation/LinksGroup.cs

```

1 using System;
2 using System.Collections.Generic;
3 using System.Runtime.CompilerServices;
4 using Platform.Collections;
5 using Platform.Collections.Lists;
6
7 namespace Link.Foundation.Links.Notation
8 {
9     /// <summary>
10     /// Represents a group of links with hierarchical structure, where each group contains a
11     ↪ primary link and optional nested groups.
12     /// This structure supports the indentation-based syntax of the Links Notation.
13     /// </summary>
14     ↪ /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
15     any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
16     public struct LinksGroup<TLinkAddress> : IEquatable<LinksGroup<TLinkAddress>>
17     {
18         /// <summary>
19         /// Gets or sets the primary link of this group.
20         /// </summary>
21         public Link<TLinkAddress> Link
22         {
23             [MethodImpl(MethodImplOptions.AggressiveInlining)]
24             get;
25             [MethodImpl(MethodImplOptions.AggressiveInlining)]
26             set;
27         }
28
29         /// <summary>
30         ↪ /// Gets or sets the collection of nested groups within this group. Can be null if no
31         nested groups exist.
32         /// </summary>
33         public IList<LinksGroup<TLinkAddress>>? Groups
34         {
35             get;
36             set;
37         }
38     }
39 }

```

```

32         [MethodImpl(MethodImplOptions.AggressiveInlining)]
33         get;
34         [MethodImpl(MethodImplOptions.AggressiveInlining)]
35         set;
36     }
37
38     /// <summary>
39     /// Gets or sets a value indicating whether this group was written as a parenthesized
↪ group.
40     /// A parenthesized group is a link in its own right, so it keeps its identifier and
↪ values
41     /// instead of being unwrapped when it is used as a child of the indented identifier
↪ syntax.
42     /// This is a parse-time hint and therefore takes no part in equality comparisons.
43     /// </summary>
44     internal bool IsParenthesized
45     {
46         [MethodImpl(MethodImplOptions.AggressiveInlining)]
47         get;
48         [MethodImpl(MethodImplOptions.AggressiveInlining)]
49         set;
50     }
51
52     /// <summary>
53     /// Initializes a new links group with the specified link and nested groups.
54     /// </summary>
55     /// <param name="link">The primary link of this group.</param>
56     /// <param name="groups">The nested groups within this group.</param>
57     [MethodImpl(MethodImplOptions.AggressiveInlining)]
58     public LinksGroup(Link<TLinkAddress> link, IList<LinksGroup<TLinkAddress>>? groups)
59     {
60         Link = link;
61         Groups = groups;
62     }
63
64     /// <summary>
65     /// Initializes a new links group with the specified link and no nested groups.
66     /// </summary>
67     /// <param name="link">The primary link of this group.</param>
68     [MethodImpl(MethodImplOptions.AggressiveInlining)]
69     public LinksGroup(Link<TLinkAddress> link) : this(link, null) { }
70
71     /// <summary>
72     /// Implicitly converts a links group to a list of links by flattening the hierarchical
↪ structure.
73     /// </summary>
74     /// <param name="value">The links group to convert.</param>
75     [MethodImpl(MethodImplOptions.AggressiveInlining)]
76     public static implicit operator List<Link<TLinkAddress>>(LinksGroup<TLinkAddress> value)
↪ => value.ToLinksList();
77
78     /// <summary>
79     /// Converts this links group to a flat list of links, preserving the hierarchical
↪ relationships.
80     /// </summary>
81     /// <returns>A list of links representing the flattened structure of this
↪ group.</returns>
82     [MethodImpl(MethodImplOptions.AggressiveInlining)]
83     public List<Link<TLinkAddress>> ToLinksList()
84     {
85         var list = new List<Link<TLinkAddress>>();
86         AppendToLinksList(list);
87         return list;
88     }
89
90     /// <summary>
91     /// Appends this links group to the specified list as flattened links.
92     /// </summary>
93     /// <param name="list">The list to append the links to.</param>
94     [MethodImpl(MethodImplOptions.AggressiveInlining)]
95     public void AppendToLinksList(List<Link<TLinkAddress>> list) => AppendToLinksList(list,
↪ Link, this);
96
97     /// <summary>
98     /// Recursively appends a links group to a list, combining dependencies with nested
↪ groups to create a flattened structure.
99     /// </summary>
100    /// <param name="list">The list to append the links to.</param>
101    /// <param name="dependency">The dependency link to combine with nested groups.</param>

```

```

102     /// <param name="group">The links group to process.</param>
103     [MethodImpl(MethodImplOptions.AggressiveInlining)]
104     public static void AppendToLinksList(List<Link<TLinkAddress>> list, Link<TLinkAddress>
↳ dependency, LinksGroup<TLinkAddress> group)
105     {
106         list.Add(dependency);
107         var groups = group.Groups;
108         if (groups != null && !groups.IsNullOrEmpty())
109         {
110             for (int i = 0; i < groups.Count; i++)
111             {
112                 var innerGroup = groups[i];
113                 AppendToLinksList(list, dependency.Combine(innerGroup.Link), innerGroup);
114             }
115         }
116     }
117
118     /// <summary>
119     /// Determines whether the specified object is equal to this links group.
120     /// </summary>
121     /// <param name="obj">The object to compare with this links group.</param>
122     /// <returns>True if the specified object is equal to this links group; otherwise,
↳ false.</returns>
123     [MethodImpl(MethodImplOptions.AggressiveInlining)]
124     public override bool Equals(object? obj) => obj is LinksGroup<TLinkAddress> linksGroup ?
↳ Equals(linksGroup) : false;
125
126     /// <summary>
127     /// Returns the hash code for this links group.
128     /// </summary>
129     /// <returns>A hash code for this links group.</returns>
130     [MethodImpl(MethodImplOptions.AggressiveInlining)]
131     public override int GetHashCode() => (Link, (Groups ??
↳ Array.Empty<LinksGroup<TLinkAddress>>()).GenerateHashCode()).GetHashCode();
132
133     /// <summary>
134     /// Indicates whether the current links group is equal to another links group.
135     /// </summary>
136     /// <param name="other">The links group to compare with this links group.</param>
137     /// <returns>True if the current links group is equal to the other parameter; otherwise,
↳ false.</returns>
138     [MethodImpl(MethodImplOptions.AggressiveInlining)]
139     public bool Equals(LinksGroup<TLinkAddress> other) => Link.Equals(other.Link) && (Groups
↳ ?? Array.Empty<LinksGroup<TLinkAddress>>()).EqualTo(other.Groups ??
↳ Array.Empty<LinksGroup<TLinkAddress>>());
140
141     /// <summary>
142     /// Determines whether two links group instances are equal.
143     /// </summary>
144     /// <param name="left">The first links group to compare.</param>
145     /// <param name="right">The second links group to compare.</param>
146     /// <returns>True if the links groups are equal; otherwise, false.</returns>
147     [MethodImpl(MethodImplOptions.AggressiveInlining)]
148     public static bool operator ==(LinksGroup<TLinkAddress> left, LinksGroup<TLinkAddress>
↳ right) => left.Equals(right);
149
150     /// <summary>
151     /// Determines whether two links group instances are not equal.
152     /// </summary>
153     /// <param name="left">The first links group to compare.</param>
154     /// <param name="right">The second links group to compare.</param>
155     /// <returns>True if the links groups are not equal; otherwise, false.</returns>
156     [MethodImpl(MethodImplOptions.AggressiveInlining)]
157     public static bool operator !=(LinksGroup<TLinkAddress> left, LinksGroup<TLinkAddress>
↳ right) => !(left == right);
158
159     /// <summary>
160     /// Returns a string representation of this links group.
161     /// </summary>
162     /// <returns>A string representation of this links group.</returns>
163     [MethodImpl(MethodImplOptions.AggressiveInlining)]
164     public override string ToString() => ToLinksList().Format();
165 }
166 }

```

1.8 ./Link.Foundation.Links.Notation/ .cs

```

1 using System.Runtime.CompilerServices;
2

```

```

3 namespace Link.Foundation.Links.Notation
4 {
5     /// <summary>
6     /// A utility struct that provides convenient implicit conversions for creating links from
7     /// various tuple combinations.
8     /// The underscore name follows functional programming conventions for placeholder/utility
9     /// types.
10    /// </summary>
11    /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
12    /// any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
13    public struct <TLinkAddress>
14    {
15        /// <summary>
16        /// Gets the underlying link represented by this utility struct.
17        /// </summary>
18        public readonly Link<TLinkAddress> Link;
19
20        /// <summary>
21        /// Initializes a new instance with the specified link.
22        /// </summary>
23        /// <param name="id">The link to wrap.</param>
24        [MethodImpl(MethodImplOptions.AggressiveInlining)]
25        public (Link<TLinkAddress> id) => Link = id;
26
27        /// <summary>
28        /// Implicitly converts a link to this utility struct.
29        /// </summary>
30        /// <param name="value">The link to convert.</param>
31        [MethodImpl(MethodImplOptions.AggressiveInlining)]
32        public static implicit operator <TLinkAddress>(Link<TLinkAddress> value) => new(value);
33
34        /// <summary>
35        /// Implicitly converts a link address to this utility struct.
36        /// </summary>
37        /// <param name="id">The link address to convert.</param>
38        [MethodImpl(MethodImplOptions.AggressiveInlining)]
39        public static implicit operator <TLinkAddress>(TLinkAddress id) => new(id);
40
41        /// <summary>
42        /// Implicitly converts a tuple of address and link to this utility struct.
43        /// </summary>
44        /// <param name="value">The tuple containing address and link.</param>
45        [MethodImpl(MethodImplOptions.AggressiveInlining)]
46        public static implicit operator <TLinkAddress>((TLinkAddress, Link<TLinkAddress>)
47        value) => new Link<TLinkAddress>(value.Item1, new[] { value.Item2 });
48
49        /// <summary>
50        /// Implicitly converts a tuple of address and two links to this utility struct.
51        /// </summary>
52        /// <param name="value">The tuple containing address and two links.</param>
53        [MethodImpl(MethodImplOptions.AggressiveInlining)]
54        public static implicit operator <TLinkAddress>((TLinkAddress, Link<TLinkAddress>,
55        Link<TLinkAddress>) value) => new Link<TLinkAddress>(value.Item1, new[] { value.Item2,
56        value.Item3 });
57
58        /// <summary>
59        /// Implicitly converts a tuple of address and three links to this utility struct.
60        /// </summary>
61        /// <param name="value">The tuple containing address and three links.</param>
62        [MethodImpl(MethodImplOptions.AggressiveInlining)]
63        public static implicit operator <TLinkAddress>((TLinkAddress, Link<TLinkAddress>,
64        Link<TLinkAddress>, Link<TLinkAddress>) value) => new Link<TLinkAddress>(value.Item1, new[]
65        { value.Item2, value.Item3, value.Item4 });
66
67        /// <summary>
68        /// Implicitly converts this utility struct back to a link.
69        /// </summary>
70        /// <param name="value">The utility struct to convert.</param>
71        [MethodImpl(MethodImplOptions.AggressiveInlining)]
72        public static implicit operator Link<TLinkAddress>(<TLinkAddress> value) => value.Link;
73    }
74 }

```

1.9 ./Link.Foundation.Links.Notation/id.cs

```

1 using System.Runtime.CompilerServices;
2
3 namespace Link.Foundation.Links.Notation
4 {
5     /// <summary>

```

```

6     /// A readonly struct that explicitly represents a link identifier/address.
7     /// This type is used to distinguish between regular values and explicit identifiers in link
    ↪ construction.
8     /// </summary>
9     /// <typeparam name="TLinkAddress">The type used for link addresses/identifiers. This can be
    ↪ any type that uniquely identifies a link, such as string, int, or Guid.</typeparam>
10    public readonly struct id<TLinkAddress>
11    {
12        /// <summary>
13        /// Gets the identifier value.
14        /// </summary>
15        public readonly TLinkAddress Id;
16
17        /// <summary>
18        /// Initializes a new identifier with the specified value.
19        /// </summary>
20        /// <param name="id">The identifier value.</param>
21        [MethodImpl(MethodImplOptions.AggressiveInlining)]
22        public id(TLinkAddress id) => Id = id;
23
24        /// <summary>
25        /// Explicitly converts a value to an identifier struct.
26        /// </summary>
27        /// <param name="id">The value to convert to an identifier.</param>
28        [MethodImpl(MethodImplOptions.AggressiveInlining)]
29        public static explicit operator id<TLinkAddress>(TLinkAddress id) => new(id);
30    }
31 }

```

1.10 ./Link.Foundation.Links.Notation.Tests/ApiTests.cs

```

1  using System;
2  using System.Collections.Generic;
3  using Xunit;
4
5  namespace Link.Foundation.Links.Notation.Tests
6  {
7      public static class ApiTests
8      {
9          [Fact]
10         public static void IsRefTest()
11         {
12             // C# doesn't have separate Ref/Link types, but we can test simple link behavior
13             var simpleLink = new Link<string>("some value", null);
14             Assert.Equal("some value", simpleLink.Id);
15             Assert.Null(simpleLink.Values);
16         }
17
18         [Fact]
19         public static void IsLinkTest()
20         {
21             // Test link with values
22             var values = new List<Link<string>> { new Link<string>("child", null) };
23             var link = new Link<string>("id", values);
24             Assert.Equal("id", link.Id);
25             Assert.Single(link.Values!);
26             Assert.Equal("child", link.Values?[0].Id);
27         }
28
29         [Fact]
30         public static void IsRefEquivalentTest()
31         {
32             // C# doesn't have separate Ref/Link types, but we can test simple link behavior
33             var simpleLink = new Link<string>("some value", null);
34             Assert.Equal("some value", simpleLink.Id);
35             Assert.Null(simpleLink.Values);
36         }
37
38         [Fact]
39         public static void IsLinkEquivalentTest()
40         {
41             // Test link with values
42             var values = new List<Link<string>> { new Link<string>("child", null) };
43             var link = new Link<string>("id", values);
44             Assert.Equal("id", link.Id);
45             Assert.Single(link.Values!);
46             Assert.Equal("child", link.Values?[0].Id);
47         }
48
49         [Fact]

```

```

50 public static void EmptyLinkTest()
51 {
52     var link = new Link<string>(null, new List<Link<string>>());
53     var output = link.ToString();
54     Assert.Equal("()", output);
55 }
56
57 [Fact]
58 public static void SimpleLinkTest()
59 {
60     var input = "(1: 1 1)";
61     var parser = new Parser();
62     var parsed = parser.Parse(input);
63
64     // Validate regular formatting
65     var output = parsed.Format();
66     Assert.Contains("1:", output);
67     Assert.Contains("1", output);
68 }
69
70 [Fact]
71 public static void LinkWithSourceTargetTest()
72 {
73     var input = "(index: source target)";
74     var parser = new Parser();
75     var parsed = parser.Parse(input);
76
77     // Validate regular formatting
78     var output = parsed.Format();
79     Assert.Equal(input, output);
80 }
81
82 [Fact]
83 public static void LinkWithSourceTypeTargetTest()
84 {
85     var input = "(index: source type target)";
86     var parser = new Parser();
87     var parsed = parser.Parse(input);
88
89     // Validate regular formatting
90     var output = parsed.Format();
91     Assert.Equal(input, output);
92 }
93
94 [Fact]
95 public static void SingleLineFormatTest()
96 {
97     var input = "id: value1 value2";
98     var parser = new Parser();
99     var parsed = parser.Parse(input);
100
101     // The parser should handle single-line format
102     var output = parsed.Format();
103     Assert.Contains("id", output);
104     Assert.Contains("value1", output);
105     Assert.Contains("value2", output);
106 }
107
108 [Fact]
109 public static void QuotedReferencesTest()
110 {
111     var input = @"("quoted id": "value with spaces")";
112     var parser = new Parser();
113     var parsed = parser.Parse(input);
114
115     var output = parsed.Format();
116     Assert.Contains("quoted id", output);
117     Assert.Contains("value with spaces", output);
118 }
119
120 [Fact]
121 public static void QuotedReferencesParsingTest()
122 {
123     // Test that quoted references are parsed correctly
124     var input = @"("quoted id": "value with spaces")";
125     var parser = new Parser();
126     var parsed = parser.Parse(input);
127
128     // Verify parsing worked correctly

```



```

129     var output = parsed.Format();
130     Assert.Contains("quoted id", output);
131     Assert.Contains("value with spaces", output);
132 }
133
134 [Fact]
135 public static void IndentedIdSyntaxParsingTest()
136 {
137     // Test that indented ID syntax is parsed correctly
138     var indented = "id:\n value1\n value2";
139     var inline = "(id: value1 value2)";
140
141     var parser = new Parser();
142     var indentedParsed = parser.Parse(indented);
143     var inlineParsed = parser.Parse(inline);
144
145     // Both should produce equivalent structures
146     var indentedOutput = indentedParsed.Format();
147     var inlineOutput = inlineParsed.Format();
148     Assert.Equal(indentedOutput, inlineOutput);
149     Assert.Equal("(id: value1 value2)", indentedOutput);
150 }
151
152 [Fact]
153 public static void IndentedIdSyntaxRoundtripTest()
154 {
155     var input = "id:\n value1\n value2";
156     var parser = new Parser();
157     var parsed = parser.Parse(input);
158
159     // Validate that we can format with indented syntax using FormatOptions
160     var options = new FormatOptions
161     {
162         MaxInlineRefs = 1, // Force indentation with more than 1 ref
163         PreferInline = false
164     };
165     var output = parsed.Format(options);
166     Assert.Equal(input, output);
167 }
168
169 [Fact]
170 public static void MultipleIndentedIdSyntaxParsingTest()
171 {
172     // Test that multiple indented ID links are parsed correctly
173     var indented = "id1:\n a\n b\nid2:\n c\n d";
174     var inline = "(id1: a b)\n(id2: c d)";
175
176     var parser = new Parser();
177     var indentedParsed = parser.Parse(indented);
178     var inlineParsed = parser.Parse(inline);
179
180     // Both should produce equivalent structures
181     var indentedOutput = indentedParsed.Format();
182     var inlineOutput = inlineParsed.Format();
183     Assert.Equal(indentedOutput, inlineOutput);
184     Assert.Equal("(id1: a b)\n(id2: c d)", indentedOutput);
185 }
186
187 [Fact]
188 public static void MultipleIndentedIdSyntaxRoundtripTest()
189 {
190     var input = "id1:\n a\n b\nid2:\n c\n d";
191     var parser = new Parser();
192     var parsed = parser.Parse(input);
193
194     // Validate that we can format with indented syntax using FormatOptions
195     var options = new FormatOptions
196     {
197         MaxInlineRefs = 1, // Force indentation with more than 1 ref
198         PreferInline = false
199     };
200     var output = parsed.Format(options);
201     Assert.Equal(input, output);
202 }
203 }
204 }

```

1.11 ./Link.Foundation.Links.Notation.Tests/EdgeCaseParserTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class EdgeCaseParserTests
7      {
8          [Fact]
9          public static void EmptyLinkTest()
10         {
11             var source = @": ";
12             var parser = new Parser();
13             // Standalone ':' is now forbidden and should throw an exception
14             Assert.Throws<FormatException>(() => parser.Parse(source));
15         }
16
17         [Fact]
18         public static void EmptyLinkWithParenthesesTest()
19         {
20             var source = @"() ";
21             var target = @"() ";
22             var parser = new Parser();
23             var links = parser.Parse(source);
24             var formattedLinks = links.Format();
25             Assert.Equal(target, formattedLinks);
26         }
27
28         [Fact]
29         public static void EmptyLinkWithEmptySelfReferenceTest()
30         {
31             var source = @"(:) ";
32             var parser = new Parser();
33             // '(:)' is now forbidden and should throw an exception
34             Assert.Throws<FormatException>(() => parser.Parse(source));
35         }
36
37         [Fact]
38         public static void AllFeaturesTest()
39         {
40             // Test single-line link with id
41             var input = "id: value1 value2";
42             var result = new Parser().Parse(input);
43             Assert.NotEmpty(result);
44
45             // Test multi-line link with id
46             input = "(id: value1 value2)";
47             result = new Parser().Parse(input);
48             Assert.NotEmpty(result);
49
50             // Test link without id (single-line) - now forbidden
51             input = ": value1 value2";
52             Assert.Throws<FormatException>(() => new Parser().Parse(input));
53
54             // Test link without id (multi-line) - now forbidden
55             input = "(: value1 value2)";
56             Assert.Throws<FormatException>(() => new Parser().Parse(input));
57
58             // Test singlet link
59             input = "(singlet)";
60             result = new Parser().Parse(input);
61             Assert.Single(result);
62             Assert.Null(result[0].Id);
63             Assert.NotNull(result[0].Values);
64             Assert.Single(result[0].Values!);
65             Assert.Equal("singlet", result[0].Values?[0].Id);
66             Assert.Null(result[0].Values?[0].Values);
67
68             // Test value link
69             input = "(value1 value2 value3)";
70             result = new Parser().Parse(input);
71             Assert.NotEmpty(result);
72
73             // Test quoted references
74             input = @"("id with spaces": "value with spaces")";
75             result = new Parser().Parse(input);
76             Assert.NotEmpty(result);
77
78             // Test single-quoted references

```

```

79     input = "('id': 'value')";
80     result = new Parser().Parse(input);
81     Assert.NotEmpty(result);
82
83     // Test nested links
84     input = "(outer: (inner: value))";
85     result = new Parser().Parse(input);
86     Assert.NotEmpty(result);
87 }
88
89 [Fact]
90 public static void TestSingletLinks()
91 {
92     // Test singlet (1)
93     var input = "(1)";
94     var result = new Parser().Parse(input);
95     Assert.Single(result);
96     Assert.Null(result[0].Id);
97     Assert.NotNull(result[0].Values);
98     Assert.Single(result[0].Values!);
99     Assert.Equal("1", result[0].Values?[0].Id);
100    Assert.Null(result[0].Values?[0].Values);
101
102    // Test (1 2)
103    input = "(1 2)";
104    result = new Parser().Parse(input);
105    Assert.Single(result);
106    Assert.Null(result[0].Id);
107    Assert.NotNull(result[0].Values);
108    Assert.Equal(2, result[0].Values?.Count);
109    Assert.Equal("1", result[0].Values?[0].Id);
110    Assert.Null(result[0].Values?[0].Values);
111    Assert.Equal("2", result[0].Values?[1].Id);
112    Assert.Null(result[0].Values?[1].Values);
113
114    // Test (1 2 3)
115    input = "(1 2 3)";
116    result = new Parser().Parse(input);
117    Assert.Single(result);
118    Assert.Null(result[0].Id);
119    Assert.NotNull(result[0].Values);
120    Assert.Equal(3, result[0].Values?.Count);
121    Assert.Equal("1", result[0].Values?[0].Id);
122    Assert.Null(result[0].Values?[0].Values);
123    Assert.Equal("2", result[0].Values?[1].Id);
124    Assert.Null(result[0].Values?[1].Values);
125    Assert.Equal("3", result[0].Values?[2].Id);
126    Assert.Null(result[0].Values?[2].Values);
127
128    // Test (1 2 3 4)
129    input = "(1 2 3 4)";
130    result = new Parser().Parse(input);
131    Assert.Single(result);
132    Assert.Null(result[0].Id);
133    Assert.NotNull(result[0].Values);
134    Assert.Equal(4, result[0].Values?.Count);
135    Assert.Equal("1", result[0].Values?[0].Id);
136    Assert.Null(result[0].Values?[0].Values);
137    Assert.Equal("2", result[0].Values?[1].Id);
138    Assert.Null(result[0].Values?[1].Values);
139    Assert.Equal("3", result[0].Values?[2].Id);
140    Assert.Null(result[0].Values?[2].Values);
141    Assert.Equal("4", result[0].Values?[3].Id);
142    Assert.Null(result[0].Values?[3].Values);
143 }
144
145 [Fact]
146 public static void EmptyDocumentTest()
147 {
148     var input = "";
149     // Empty document should return empty list
150     var result = new Parser().Parse(input);
151     Assert.Empty(result);
152 }
153
154 [Fact]
155 public static void WhitespaceOnlyTest()
156 {

```

```

157     var input = " \n \n ";
158     // Whitespace-only document should return empty list (similar to empty document)
159     var result = new Parser().Parse(input);
160     Assert.Empty(result);
161 }
162
163 [Fact]
164 public static void EmptyLinksTest()
165 {
166     var input = "()";
167     var result = new Parser().Parse(input);
168     Assert.NotEmpty(result);
169
170     // '(:)' is now forbidden
171     input = "(:)";
172     Assert.Throws<FormatException>(() => new Parser().Parse(input));
173
174     input = "(id:)";
175     result = new Parser().Parse(input);
176     Assert.NotEmpty(result);
177 }
178
179 [Fact]
180 public static void InvalidInputTest()
181 {
182     var input = "(invalid";
183     // Unclosed parentheses should throw an exception
184     Assert.Throws<FormatException>(() => new Parser().Parse(input));
185 }
186 }
187 }

```

1.12 ./Link.Foundation.Links.Notation.Tests/EmptyReferenceTests.cs

```

1 using System.Collections.Generic;
2 using System.Linq;
3 using Xunit;
4
5 namespace Link.Foundation.Links.Notation.Tests
6 {
7     /// <summary>
8     /// Conformance tests for the empty reference.
9     ///
10    /// https://github.com/link-foundation/links-notation/issues/288
11    ///
12    /// A bare delimiter pair is the empty reference. The three delimiters
13    /// <c>"</c>, <c>'</c> and <c>`</c> behave identically, and every longer
14    /// n-quote run keeps the meaning it already had. The table below is shared
15    /// with the Rust, JavaScript, Python, Go, Java and PHP suites, so a document
16    /// written by one implementation reads the same in all of them.
17    /// </summary>
18    public static class EmptyReferenceTests
19    {
20        /// <summary>
21        /// Renders a parsed node unambiguously: every reference is wrapped in
22        /// angle brackets so an empty one is visible as < >.
23        /// </summary>
24        private static string Render(Link<string> node)
25        {
26            if (node.Values == null || node.Values.Count == 0)
27            {
28                return "<" + (node.Id ?? "") + ">";
29            }
30            var head = node.Id == null ? "" : "<" + node.Id + ">: ";
31            return "(" + head + string.Join(" ", node.Values.Select(Render)) + ")";
32        }
33
34        private static string Rendered(string source)
35        {
36            var links = new Parser().Parse(source);
37            return string.Join("\n", links.Select(Render));
38        }
39
40        private static void AssertParsesAs(string source, string expected)
41        {
42            Assert.Equal(expected, Rendered(source));
43        }
44
45        [Fact]
46        public static void BareDelimiterPairIsTheEmptyReference()

```

```

47 {
48     AssertParsesAs("(a \" b)", "<a> <> <b>");
49 }
50
51 [Fact]
52 public static void EveryDelimiterStyleYieldsTheSameEmptyReference()
53 {
54     AssertParsesAs("(a \" b)", "<a> <> <b>");
55     AssertParsesAs("(a ' b)", "<a> <> <b>");
56     AssertParsesAs("(a ` b)", "<a> <> <b>");
57 }
58
59 [Fact]
60 public static void AdjacentEmptyReferencesStaySeparate()
61 {
62     AssertParsesAs("(a \" \" b)", "<a> <> <> <b>");
63     AssertParsesAs("(a ' ' b)", "<a> <> <> <b>");
64     AssertParsesAs("(a ` ` b)", "<a> <> <> <b>");
65     AssertParsesAs("(a \" \" b)", "<a> <> <> <b>");
66 }
67
68 [Fact]
69 public static void NestedEmptyReferencesParse()
70 {
71     AssertParsesAs("(\" \" (\" \" 1))", "<> (<> <1>)");
72     AssertParsesAs("(\" \" (' 1))", "<> (<> <1>)");
73     AssertParsesAs("(\"x\" (\" \" 1))", "<x> (<> <1>)");
74     AssertParsesAs("(\"x\" (\"x\" 1))", "<> (<x> <1>)");
75     AssertParsesAs("(\" \" x (\" \" 1))", "<> <x> (<> <1>)");
76     AssertParsesAs("(\" \" 1 (\" \" 1))", "<> <1> (<> <1>)");
77 }
78
79 [Fact]
80 public static void EmptyReferenceIsValidAsAnId()
81 {
82     AssertParsesAs("(\" \": 1)", "<>: <1>");
83     AssertParsesAs("(o: (\" \" (o: (\" \" 1))))", "<o>: (<> (<o>: (<> <1>)))");
84 }
85
86 [Fact]
87 public static void NQuoteDelimitedBodiesAreUnchanged()
88 {
89     // A run that encloses a substantive body keeps its n-quote meaning.
90     AssertParsesAs("(a \" \"x\" \" b)", "<a> <x> <b>");
91     AssertParsesAs("(x \" \" \" \" \")", "<x> < \" >");
92     AssertParsesAs("(x ' \" ')", "<x> < \" >");
93     // An n-quote-delimited empty is still empty.
94     AssertParsesAs("(a \" \" \" \" b)", "<a> <> <b>");
95 }
96
97 [Fact]
98 public static void ASingleSpaceStillReadsAsASpace()
99 {
100     AssertParsesAs("(a \" \" b)", "<a> < > <b>");
101 }
102
103 [Fact]
104 public static void EmptyReferenceSurvivesARoundTrip()
105 {
106     var sources = new[]
107     {
108         "(a \" \" b)",
109         "(a \" \" \" \" b)",
110         "(\" \" (\" \" 1))",
111         "(\" \": 1)",
112         "(o: (\" \" (o: (\" \" 1))))",
113     };
114     foreach (var source in sources)
115     {
116         var formatted = ((IList<Link<string>>)new Parser().Parse(source)).Format();
117         var reparsed = ((IList<Link<string>>)new Parser().Parse(formatted)).Format();
118         Assert.Equal(formatted, reparsed);
119     }
120 }
121
122 [Fact]
123 public static void EmptyReferenceIsWrittenAsADelimiterPair()
124 {

```

```

125         var links = (IList<Link<string>>)new Parser().Parse("(a \\\" b)");
126         Assert.Equal("(a \\\" b)", links.Format());
127     }
128 }
129 }

```

1.13 ./Link.Foundation.Links.Notation.Tests/FormatConfigTests.cs

```

1  using Xunit;
2
3  namespace Link.Foundation.Links.Notation.Tests
4  {
5      public class FormatConfigTests
6      {
7          [Fact]
8          public void FormatConfigBasicTest()
9          {
10             var config = new FormatConfig();
11             Assert.False(config.LessParentheses);
12             Assert.Equal(80, config.MaxLineLength);
13             Assert.False(config.IndentLongLines);
14         }
15
16         [Fact]
17         public void FormatWithLineLengthLimitTest()
18         {
19             // Create a config with line length limit
20             // The line would be 32+ chars, so use threshold of 30
21             var config = new FormatConfig
22             {
23                 IndentLongLines = true,
24                 MaxLineLength = 30,
25                 PreferInline = false
26             };
27
28             // Verify config is set correctly
29             Assert.True(config.IndentLongLines);
30             Assert.Equal(30, config.MaxLineLength);
31             // Note: Full formatting integration would test actual output here
32         }
33
34         [Fact]
35         public void FormatWithMaxInlineRefsTest()
36         {
37             // Create a config with MaxInlineRefs=3 (should trigger indentation with 4 refs)
38             var config = new FormatConfig
39             {
40                 MaxInlineRefs = 3,
41                 PreferInline = false
42             };
43
44             // Verify config is set correctly
45             Assert.Equal(3, config.MaxInlineRefs);
46             Assert.True(config.ShouldIndentByRefCount(4));
47             // Note: Full formatting integration would test actual output here
48         }
49
50         [Fact]
51         public void FormatWithConsecutiveGroupingTest()
52         {
53             // Create a config with consecutive grouping enabled
54             var config = new FormatConfig
55             {
56                 GroupConsecutive = true
57             };
58
59             // Verify config is set correctly
60             Assert.True(config.GroupConsecutive);
61             // Note: Full formatting integration would test grouping behavior here
62         }
63
64         [Fact]
65         public void FormatConfigCustomIndentTest()
66         {
67             var config = new FormatConfig
68             {
69                 MaxInlineRefs = 3,
70                 PreferInline = false,
71                 IndentString = "    " // 4 spaces
72             };
73

```

```

74     Assert.Equal("    ", config.IndentString);
75 }
76
77 [Fact]
78 public void FormatConfigLessParenthesesTest()
79 {
80     var config = new FormatConfig
81     {
82         LessParentheses = true
83     };
84
85     Assert.True(config.LessParentheses);
86 }
87
88 [Fact]
89 public void RoundtripWithLineLengthFormattingTest()
90 {
91     // This test would require full format integration
92     // For now, just verify config creation
93     var config = new FormatConfig
94     {
95         MaxInlineRefs = 2,
96         PreferInline = false
97     };
98
99     Assert.Equal(2, config.MaxInlineRefs);
100 }
101
102 [Fact]
103 public void ShouldIndentByLengthTest()
104 {
105     var config = new FormatConfig
106     {
107         IndentLongLines = true,
108         MaxLineLength = 80
109     };
110
111     Assert.False(config.ShouldIndentByLength("short"));
112     Assert.True(config.ShouldIndentByLength(new string('a', 100)));
113 }
114
115 [Fact]
116 public void ShouldIndentByRefCountTest()
117 {
118     var config = new FormatConfig
119     {
120         MaxInlineRefs = 3
121     };
122
123     Assert.False(config.ShouldIndentByRefCount(2));
124     Assert.False(config.ShouldIndentByRefCount(3));
125     Assert.True(config.ShouldIndentByRefCount(4));
126 }
127 }
128 }

```

1.14 ./Link.Foundation.Links.Notation.Tests/IndentationConsistencyTests.cs

```

1  using Xunit;
2  using System.Collections.Generic;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public class IndentationConsistencyTests
7      {
8          [Fact]
9          public void LeadingSpacesVsNoLeadingSpaces()
10         {
11             // Example with 2 leading spaces (from issue #135)
12             var withLeading = @"    TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
13 TELEGRAM ALLOWED CHATS:
14     -1002975819706
15     -1002861722681
16 TELEGRAM HIVE OVERRIDES:
17     --all-issues
18     --once
19 TELEGRAM BOT VERBOSE: true";
20
21             // Example without leading spaces (from issue #135)
22             var withoutLeading = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
23 TELEGRAM ALLOWED CHATS:
24     -1002975819706

```

```

25     -1002861722681
26     TELEGRAM HIVE OVERRIDES:
27         --all-issues
28         --once
29     TELEGRAM BOT VERBOSE: true";
30
31     var resultWith = new Parser().Parse(withLeading);
32     var resultWithout = new Parser().Parse(withoutLeading);
33
34     // Compare the entire formatted output (complete round trip test)
35     Assert.Equal(resultWithout.Format(), resultWith.Format());
36 }
37
38 [Fact]
39 public void TwoSpacesVsFourSpacesIndentation()
40 {
41     // Example with 2 spaces per level
42     var twoSpaces = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
TELEGRAM ALLOWED CHATS:
    -1002975819706
    -1002861722681
TELEGRAM HIVE OVERRIDES:
    --all-issues
    --once
    --auto-fork
    --skip-issues-with-prs
    --attach-logs
    --verbose
    --no-tool-check
TELEGRAM SOLVE OVERRIDES:
    --auto-fork
    --auto-continue
    --attach-logs
    --verbose
    --no-tool-check
TELEGRAM BOT VERBOSE: true";
43
44     // Example with 4 spaces per level
45     var fourSpaces = @"TELEGRAM BOT TOKEN: '849...355:AAG...rgk YZk...aPU'
TELEGRAM ALLOWED CHATS:
    -1002975819706
    -1002861722681
TELEGRAM HIVE OVERRIDES:
    --all-issues
    --once
    --auto-fork
    --skip-issues-with-prs
    --attach-logs
    --verbose
    --no-tool-check
TELEGRAM SOLVE OVERRIDES:
    --auto-fork
    --auto-continue
    --attach-logs
    --verbose
    --no-tool-check
TELEGRAM BOT VERBOSE: true";
46
47     var resultTwo = new Parser().Parse(twoSpaces);
48     var resultFour = new Parser().Parse(fourSpaces);
49
50     // Compare the entire formatted output (complete round trip test)
51     Assert.Equal(resultFour.Format(), resultTwo.Format());
52 }
53
54 [Fact]
55 public void SimpleTwoVsFourSpacesIndentation()
56 {
57     // Simple example with 2 spaces
58     var twoSpaces = @"parent:
59     child1
60     child2";
61
62     // Simple example with 4 spaces
63     var fourSpaces = @"parent:
64     child1
65     child2";
66
67     var resultTwo = new Parser().Parse(twoSpaces);
68     var resultFour = new Parser().Parse(fourSpaces);
69
70     // Compare the entire formatted output (complete round trip test)
71     Assert.Equal(resultFour.Format(), resultTwo.Format());
72 }

```



```

108     }
109
110     [Fact]
111     public void ThreeLevelNestingWithDifferentIndentation()
112     {
113         // Three levels with 2 spaces
114         var twoSpaces = @"level1:
115 level2:
116     level3a
117     level3b
118 level2b";
119
120         // Three levels with 4 spaces
121         var fourSpaces = @"level1:
122 level2:
123     level3a
124     level3b
125 level2b";
126
127         var resultTwo = new Parser().Parse(twoSpaces);
128         var resultFour = new Parser().Parse(fourSpaces);
129
130         // Compare the entire formatted output (complete round trip test)
131         Assert.Equal(resultFour.Format(), resultTwo.Format());
132     }
133 }
134 }

```

1.15 ./Link.Foundation.Links.Notation.Tests/IndentedIdSyntaxTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class IndentedIdSyntaxTests
7      {
8          [Fact]
9          public static void BasicIndentedIdSyntaxTest()
10         {
11             var indentedSyntax = @"3:
12 papa
13 loves
14 mama";
15
16             var inlineSyntax = "(3: papa loves mama)";
17
18             var parser = new Parser();
19             var indentedResult = parser.Parse(indentedSyntax);
20             var inlineResult = parser.Parse(inlineSyntax);
21
22             var indentedFormatted = indentedResult.Format();
23             var inlineFormatted = inlineResult.Format();
24
25             Assert.Equal(inlineFormatted, indentedFormatted);
26             Assert.Equal("(3: papa loves mama)", indentedFormatted);
27         }
28
29         [Fact]
30         public static void IndentedIdSyntaxWithSingleValueTest()
31         {
32             var input = @"greeting:
33 hello";
34
35             var parser = new Parser();
36             var result = parser.Parse(input);
37             var formatted = result.Format();
38
39             Assert.Equal("(greeting: hello)", formatted);
40             Assert.Single(result);
41             Assert.Equal("greeting", result[0].Id);
42             Assert.Single(result[0].Values!);
43             Assert.Equal("hello", result[0].Values![0].Id);
44         }
45
46         [Fact]
47         public static void IndentedIdSyntaxWithMultipleValuesTest()
48         {
49             var input = @"action:
50 run
51 fast
52 now";

```

```

53     var parser = new Parser();
54     var result = parser.Parse(input);
55     var formatted = result.Format();
56
57     Assert.Equal("(action: run fast now)", formatted);
58     Assert.Single(result);
59     Assert.Equal("action", result[0].Id);
60     Assert.Equal(3, result[0].Values!.Count);
61 }
62
63 [Fact]
64 public static void IndentedIdSyntaxWithNumericIdTest()
65 {
66     var input = @"42:
67 answer
68 to
69 everything";
70
71     var parser = new Parser();
72     var result = parser.Parse(input);
73     var formatted = result.Format();
74
75     Assert.Equal("(42: answer to everything)", formatted);
76 }
77
78 [Fact]
79 public static void IndentedIdSyntaxWithQuotedIdTest()
80 {
81     var input = @"""complex id"":
82 value1
83 value2";
84
85     var parser = new Parser();
86     var result = parser.Parse(input);
87     var formatted = result.Format();
88
89     Assert.Equal("( 'complex id': value1 value2)", formatted);
90 }
91
92 [Fact]
93 public static void MultipleIndentedIdLinksTest()
94 {
95     var input = @"first:
96 a
97 b
98 second:
99 c
100 d";
101
102     var parser = new Parser();
103     var result = parser.Parse(input);
104     var formatted = result.Format();
105
106     Assert.Equal(2, result.Count);
107     Assert.Contains("(first: a b)", formatted);
108     Assert.Contains("(second: c d)", formatted);
109 }
110
111 [Fact]
112 public static void MixedIndentedAndRegularSyntaxTest()
113 {
114     var input = @"first:
115 a
116 b
117 (second: c d)
118 third value";
119
120     var parser = new Parser();
121     var result = parser.Parse(input);
122
123     Assert.Equal(3, result.Count);
124
125     var formatted = result.Format();
126     Assert.Contains("(first: a b)", formatted);
127     Assert.Contains("(second: c d)", formatted);
128     Assert.Contains("third value", formatted);
129 }
130
131 [Fact]
132 public static void UnsupportedColonOnlySyntaxShouldFailTest()

```

```

133     {
134         var input = @"
135         papa
136         loves
137         mama";
138
139         var parser = new Parser();
140         Assert.Throws<FormatException>(() => parser.Parse(input));
141     }
142
143     [Fact]
144     public static void EmptyIndentedIdShouldWorkTest()
145     {
146         var input = "empty:";
147
148         var parser = new Parser();
149         var result = parser.Parse(input);
150
151         Assert.Single(result);
152         Assert.Equal("empty", result[0].Id);
153         var emptyValues = result[0].Values;
154         Assert.True(emptyValues == null || emptyValues.Count == 0);
155
156         var formatted = result.Format();
157         Assert.Equal("(empty)", formatted);
158     }
159
160     [Fact]
161     public static void EquivalenceTestComprehensiveTest()
162     {
163         var testCases = new[]
164         {
165             new { Indented = "test:\n one", Inline = "(test: one)" },
166             new { Indented = "x:\n a\n b\n c", Inline = "(x: a b c)" },
167             new { Indented = "\"quoted\":\n value", Inline = "\"(quoted\": value)" }
168         };
169
170         var parser = new Parser();
171
172         foreach (var testCase in testCases)
173         {
174             var indentedResult = parser.Parse(testCase.Indented);
175             var inlineResult = parser.Parse(testCase.Inline);
176
177             var indentedFormatted = indentedResult.Format();
178             var inlineFormatted = inlineResult.Format();
179
180             Assert.Equal(inlineFormatted, indentedFormatted);
181         }
182     }
183
184     [Fact]
185     public static void IndentedIdWithDeeperNestingTest()
186     {
187         var input = @"root:
188         child1
189         child2
190         grandchild";
191
192         var parser = new Parser();
193         var result = parser.Parse(input);
194
195         Assert.NotEmpty(result);
196
197         var rootLink = result[0];
198         Assert.Equal("root", rootLink.Id);
199         Assert.Equal(2, rootLink.Values!.Count);
200     }
201 }
202

```

1.16 ./Link.Foundation.Links.Notation.Tests/LinkTests.cs

```

1 using System;
2 using Xunit;
3 using System.Collections.Generic;
4
5 namespace Link.Foundation.Links.Notation.Tests
6 {
7     public static class LinkTests
8     {

```

```

9      [Fact]
10     public static void LinkConstructorWithIdOnlyTest()
11     {
12         var link = new Link<string>("test", null);
13         Assert.Equal("test", link.Id);
14         Assert.Null(link.Values);
15     }
16
17     [Fact]
18     public static void LinkConstructorWithIdAndValuesTest()
19     {
20         var values = new List<Link<string>> { new Link<string>("value1"), new
↪ Link<string>("value2") };
21         var link = new Link<string>("parent", values);
22         Assert.Equal("parent", link.Id);
23         Assert.Equal(2, link.Values?.Count);
24     }
25
26     [Fact]
27     public static void LinkToStringWithIdOnlyTest()
28     {
29         var link = new Link<string>("test");
30         Assert.Equal("(test)", link.ToString());
31     }
32
33     [Fact]
34     public static void LinkToStringWithValuesOnlyTest()
35     {
36         var values = new List<Link<string>> { new Link<string>("value1"), new
↪ Link<string>("value2") };
37         var link = new Link<string>(null, values);
38         Assert.Equal("(value1 value2)", link.ToString());
39     }
40
41     [Fact]
42     public static void LinkToStringWithIdAndValuesTest()
43     {
44         var values = new List<Link<string>> { new Link<string>("child1"), new
↪ Link<string>("child2") };
45         var link = new Link<string>("parent", values);
46         Assert.Equal("(parent: child1 child2)", link.ToString());
47     }
48
49     [Fact]
50     public static void LinkEqualsTest()
51     {
52         var link1 = new Link<string>("test");
53         var link2 = new Link<string>("test");
54         var link3 = new Link<string>("other");
55
56         Assert.Equal(link1, link2);
57         Assert.NotEqual(link1, link3);
58     }
59
60     [Fact]
61     public static void LinkCombineTest()
62     {
63         var link1 = new Link<string>("first");
64         var link2 = new Link<string>("second");
65         var combined = link1.Combine(link2);
66
67         Assert.Null(combined.Id);
68         Assert.Equal(2, combined.Values?.Count);
69         Assert.Equal("first", combined.Values?[0].Id);
70         Assert.Equal("second", combined.Values?[1].Id);
71     }
72
73     [Fact]
74     public static void LinkEscapeReferenceSimpleTest()
75     {
76         Assert.Equal("test", Link<string>.EscapeReference("test"));
77         Assert.Equal("test123", Link<string>.EscapeReference("test123"));
78     }
79
80     [Fact]
81     public static void LinkEscapeReferenceWithSpecialCharactersTest()
82     {
83         Assert.Equal("'has space'", Link<string>.EscapeReference("has space"));

```

```

84     Assert.Equal("'has:colon'", Link<string>.EscapeReference("has:colon"));
85     Assert.Equal("'has(paren)'", Link<string>.EscapeReference("has(paren)"));
86     Assert.Equal("'has)paren'", Link<string>.EscapeReference("has)paren"));
87     Assert.Equal("\"has'quote\"", Link<string>.EscapeReference("has'quote"));
88     Assert.Equal("'has\\quote'", Link<string>.EscapeReference("has\\quote"));
89 }
90
91 [Fact]
92 public static void LinkSimplifyTest()
93 {
94     // Test simplify with empty values
95     var link1 = new Link<string>("test", null);
96     var simplified1 = link1.Simplify();
97     Assert.Equal(link1, simplified1);
98
99     // Test simplify with single value
100    var link2 = new Link<string>(null, new List<Link<string>> { new
↪ Link<string>("single", null) });
101    var simplified2 = link2.Simplify();
102    Assert.Equal("single", simplified2.Id);
103    Assert.Null(simplified2.Values);
104 }
105 }
106 }

```

1.17 ./Link.Foundation.Links.Notation.Tests/LinksGroupTests.cs

```

1  using System;
2  using System.Collections.Generic;
3  using Xunit;
4
5  namespace Link.Foundation.Links.Notation.Tests
6  {
7      public static class LinksGroupTests
8      {
9          [Fact]
10         public static void LinksGroupConstructorTest()
11         {
12             var element = new Link<string>("root", null);
13             var groups = new List<LinksGroup<string>>
14             {
15                 new LinksGroup<string>(new Link<string>("child1", null)),
16                 new LinksGroup<string>(new Link<string>("child2", null))
17             };
18             var group = new LinksGroup<string>(element, groups);
19
20             Assert.Equal(element, group.Link);
21             Assert.Equal(groups, group.Groups);
22         }
23
24         [Fact]
25         public static void LinksGroupToListFlattensStructureTest()
26         {
27             var root = new Link<string>("root", null);
28             var child1 = new Link<string>("child1", null);
29             var child2 = new Link<string>("child2", null);
30             var grandchild = new Link<string>("grandchild", null);
31
32             var childGroup = new LinksGroup<string>(child2, new List<LinksGroup<string>>
33             {
34                 new LinksGroup<string>(grandchild)
35             });
36
37             var group = new LinksGroup<string>(root, new List<LinksGroup<string>>
38             {
39                 new LinksGroup<string>(child1),
40                 childGroup
41             });
42
43             var list = group.ToLinksList();
44             // The C# implementation creates a hierarchical structure
45             // root, root.Combine(child1), root.Combine(child2),
↪ root.Combine(child2).Combine(grandchild)
46             Assert.Equal(4, list.Count);
47             Assert.Equal(root, list[0]);
48         }
49
50         [Fact]
51         public static void LinksGroupAppendToLinksListTest()
52         {

```

```

53     var element = new Link<string>("root", null);
54     var children = new List<LinksGroup<string>>
55     {
56         new LinksGroup<string>(new Link<string>("child1", null)),
57         new LinksGroup<string>(new Link<string>("child2", null))
58     };
59     var group = new LinksGroup<string>(element, children);
60
61     var list = new List<Link<string>>();
62     group.AppendToLinksList(list);
63
64     Assert.Equal(3, list.Count);
65     Assert.Equal(element, list[0]);
66 }
67
68 [Fact]
69 public static void LinksGroupToStringTest()
70 {
71     var element = new Link<string>("root", null);
72     var children = new List<LinksGroup<string>>
73     {
74         new LinksGroup<string>(new Link<string>("child1", null)),
75         new LinksGroup<string>(new Link<string>("child2", null))
76     };
77     var group = new LinksGroup<string>(element, children);
78
79     var str = group.ToString();
80     Assert.Contains("root", str);
81     Assert.Contains("child1", str);
82     Assert.Contains("child2", str);
83 }
84
85 [Fact]
86 public static void LinksGroupConstructorEquivalentTest()
87 {
88     // Test creating a LinksGroup structure in an equivalent way
89     var root = new Link<string>("root", null);
90     var children = new List<Link<string>>
91     {
92         new Link<string>("child1", null),
93         new Link<string>("child2", null)
94     };
95
96     // Create a group with an id
97     var groupElement = new Link<string>("group", null);
98     var childGroups = new List<LinksGroup<string>>
99     {
100         new LinksGroup<string>(root),
101         new LinksGroup<string>(children[0]),
102         new LinksGroup<string>(children[1])
103     };
104     var group = new LinksGroup<string>(groupElement, childGroups);
105
106     Assert.Equal("group", group.Link.Id);
107     Assert.Equal(3, group.Groups!.Count);
108     Assert.Equal(root, group.Groups[0].Link);
109 }
110 }
111 }

```

1.18 ./Link.Foundation.Links.Notation.Tests/MixedIndentationModesTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MixedIndentationModesTests
7      {
8          [Fact]
9          public static void HeroExampleMixedModesTest()
10         {
11             var input = @"empInfo
12 employees:
13 (
14     name (James Kirk)
15     age 40
16 )
17 (
18     name (Jean-Luc Picard)

```

```

19     age 45
20 )
21 (
22     name (Wesley Crusher)
23     age 27
24 )";
25
26     var parser = new Parser();
27     var result = parser.Parse(input);
28
29     Assert.NotEmpty(result);
30     var formatted = result.Format();
31     Assert.Contains("empInfo", formatted);
32     Assert.Contains("employees", formatted);
33     Assert.Contains("James Kirk", formatted);
34     Assert.Contains("Jean-Luc Picard", formatted);
35     Assert.Contains("Wesley Crusher", formatted);
36 }
37
38 [Fact]
39 public static void HeroExampleAlternativeFormatTest()
40 {
41     var input = @"empInfo
42 (
43     employees:
44     (
45         name (James Kirk)
46         age 40
47     )
48     (
49         name (Jean-Luc Picard)
50         age 45
51     )
52     (
53         name (Wesley Crusher)
54         age 27
55     )
56 )";
57
58     var parser = new Parser();
59     var result = parser.Parse(input);
60
61     Assert.NotEmpty(result);
62     var formatted = result.Format();
63     Assert.Contains("empInfo", formatted);
64     Assert.Contains("employees:", formatted);
65     Assert.Contains("James Kirk", formatted);
66     Assert.Contains("Jean-Luc Picard", formatted);
67     Assert.Contains("Wesley Crusher", formatted);
68 }
69
70 [Fact]
71 public static void HeroExampleEquivalenceTest()
72 {
73     var version1 = @"empInfo
74 employees:
75 (
76     name (James Kirk)
77     age 40
78 )
79 (
80     name (Jean-Luc Picard)
81     age 45
82 )
83 (
84     name (Wesley Crusher)
85     age 27
86 )";
87
88     var version2 = @"empInfo
89 (
90     employees:
91     (
92         name (James Kirk)
93         age 40
94     )
95     (
96         name (Jean-Luc Picard)
97         age 45

```

```

98     )
99     (
100     name (Wesley Crusher)
101     age 27
102     )
103 );
104
105     var parser = new Parser();
106     var result1 = parser.Parse(version1);
107     var result2 = parser.Parse(version2);
108
109     Assert.NotEmpty(result1);
110     Assert.NotEmpty(result2);
111
112     var formatted1 = result1.Format();
113     var formatted2 = result2.Format();
114
115     Assert.Equal(formatted1, formatted2);
116 }
117
118 [Fact]
119 public static void SetContextWithoutColonTest()
120 {
121     var input = @"empInfo
122 employees";
123
124     var parser = new Parser();
125     var result = parser.Parse(input);
126
127     Assert.NotEmpty(result);
128     var formatted = result.Format();
129     Assert.Contains("empInfo", formatted);
130     Assert.Contains("employees", formatted);
131 }
132
133 [Fact]
134 public static void SequenceContextWithColonTest()
135 {
136     var input = @"employees:
137 James Kirk
138 Jean-Luc Picard
139 Wesley Crusher";
140
141     var parser = new Parser();
142     var result = parser.Parse(input);
143
144     Assert.NotEmpty(result);
145     Assert.Single(result);
146     var formatted = result.Format();
147     Assert.Contains("employees:", formatted);
148     Assert.Contains("James Kirk", formatted);
149     Assert.Contains("Jean-Luc Picard", formatted);
150     Assert.Contains("Wesley Crusher", formatted);
151 }
152
153 [Fact]
154 public static void SequenceContextWithComplexValuesTest()
155 {
156     var input = @"employees:
157 (
158     name (James Kirk)
159     age 40
160 )
161 (
162     name (Jean-Luc Picard)
163     age 45
164 )";
165
166     var parser = new Parser();
167     var result = parser.Parse(input);
168
169     Assert.NotEmpty(result);
170     Assert.Single(result);
171     var formatted = result.Format();
172     Assert.Contains("employees:", formatted);
173     Assert.Contains("James Kirk", formatted);
174     Assert.Contains("Jean-Luc Picard", formatted);
175 }
176
177 [Fact]

```



```

178     public static void NestedSetAndSequenceContextsTest()
179     {
180         var input = @"company
181 departments:
182     engineering
183     sales
184 employees:
185     (name John)
186     (name Jane)";
187
188         var parser = new Parser();
189         var result = parser.Parse(input);
190
191         Assert.NotEmpty(result);
192         var formatted = result.Format();
193         Assert.Contains("company", formatted);
194         Assert.Contains("departments:", formatted);
195         Assert.Contains("employees:", formatted);
196     }
197
198     [Fact]
199     public static void DeeplyNestedMixedModesTest()
200     {
201         var input = @"root
202 level1
203     level2:
204         value1
205         value2
206     level2b
207     level3";
208
209         var parser = new Parser();
210         var result = parser.Parse(input);
211
212         Assert.NotEmpty(result);
213         var formatted = result.Format();
214         Assert.Contains("root", formatted);
215         Assert.Contains("level2:", formatted);
216     }
217 }
218 }

```

1.19 ./Link.Foundation.Links.Notation.Tests/MultiQuoteParserTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MultiQuoteParserTests
7      {
8          // Helper to extract single reference ID
9          private static string? GetSingleRefId(System.Collections.Generic.IList<Link<string>>
↪ result)
10         {
11             if (result.Count == 1 && result[0].Id == null && result[0].Values?.Count == 1)
12             {
13                 return result[0].Values![0].Id;
14             }
15             return result.Count == 1 ? result[0].Id : null;
16         }
17
18         // =====
19         // Backtick Quote Tests (Single Backtick)
20         // =====
21
22         [Fact]
23         public static void TestBacktickQuotedReference()
24         {
25             var parser = new Parser();
26             var result = parser.Parse("`backtick quoted`");
27             Assert.Equal("backtick quoted", GetSingleRefId(result));
28         }
29
30         [Fact]
31         public static void TestBacktickQuotedWithSpaces()
32         {
33             var parser = new Parser();
34             var result = parser.Parse("`text with spaces`");
35             Assert.Equal("text with spaces", GetSingleRefId(result));
36         }
37     }
38 }

```

```

37
38 [Fact]
39 public static void TestBacktickQuotedMultiline()
40 {
41     var parser = new Parser();
42     var result = parser.Parse("`line1\nline2`");
43     Assert.Single(result);
44     Assert.NotNull(result[0].Values);
45     Assert.Single(result[0].Values!);
46     Assert.Equal("line1\nline2", result[0].Values![0].Id);
47 }
48
49 [Fact]
50 public static void TestBacktickQuotedWithEscapedBacktick()
51 {
52     var parser = new Parser();
53     var result = parser.Parse("`text with `` escaped backtick`");
54     Assert.Equal("text with ` escaped backtick", GetSingleRefId(result));
55 }
56
57 // =====
58 // Single Quote Tests (with escaping)
59 // =====
60
61 [Fact]
62 public static void TestSingleQuoteWithEscapedSingleQuote()
63 {
64     var parser = new Parser();
65     var result = parser.Parse("'text with ' escaped quote'");
66     Assert.Equal("text with ' escaped quote", GetSingleRefId(result));
67 }
68
69 // =====
70 // Double Quote Tests (with escaping)
71 // =====
72
73 [Fact]
74 public static void TestDoubleQuoteWithEscapedDoubleQuote()
75 {
76     var parser = new Parser();
77     var result = parser.Parse("\"text with \" escaped quote\"");
78     Assert.Equal("text with \" escaped quote", GetSingleRefId(result));
79 }
80
81 // =====
82 // Double Quotes (2 quote chars) Tests
83 // =====
84
85 [Fact]
86 public static void TestDoubleDoubleQuotes()
87 {
88     var parser = new Parser();
89     var result = parser.Parse("\"\"double double quotes\"");
90     Assert.Equal("double double quotes", GetSingleRefId(result));
91 }
92
93 [Fact]
94 public static void TestDoubleDoubleQuotesWithSingleQuoteInside()
95 {
96     var parser = new Parser();
97     var result = parser.Parse("\"\"text with \" inside\"");
98     Assert.Equal("text with \" inside", GetSingleRefId(result));
99 }
100
101 [Fact]
102 public static void TestDoubleDoubleQuotesWithEscape()
103 {
104     var parser = new Parser();
105     var result = parser.Parse("\"\"text with \"\"\" escaped double\"");
106     Assert.Equal("text with \"\" escaped double", GetSingleRefId(result));
107 }
108
109 [Fact]
110 public static void TestDoubleSingleQuotes()
111 {
112     var parser = new Parser();
113     var result = parser.Parse("'double single quotes'");
114     Assert.Equal("double single quotes", GetSingleRefId(result));

```

```

115     }
116
117     [Fact]
118     public static void TestDoubleSingleQuotesWithSingleQuoteInside()
119     {
120         var parser = new Parser();
121         var result = parser.Parse("'text with ' inside'");
122         Assert.Equal("text with ' inside", GetSingleRefId(result));
123     }
124
125     [Fact]
126     public static void TestDoubleSingleQuotesWithEscape()
127     {
128         var parser = new Parser();
129         var result = parser.Parse("'text with ''' escaped single'");
130         Assert.Equal("text with ' escaped single", GetSingleRefId(result));
131     }
132
133     [Fact]
134     public static void TestDoubleBacktickQuotes()
135     {
136         var parser = new Parser();
137         var result = parser.Parse("`double backtick quotes`");
138         Assert.Equal("double backtick quotes", GetSingleRefId(result));
139     }
140
141     [Fact]
142     public static void TestDoubleBacktickQuotesWithBacktickInside()
143     {
144         var parser = new Parser();
145         var result = parser.Parse("`text with ` inside`");
146         Assert.Equal("text with ` inside", GetSingleRefId(result));
147     }
148
149     [Fact]
150     public static void TestDoubleBacktickQuotesWithEscape()
151     {
152         var parser = new Parser();
153         var result = parser.Parse("`text with ``` escaped backtick`");
154         Assert.Equal("text with `` escaped backtick", GetSingleRefId(result));
155     }
156
157     // =====
158     // Triple Quotes (3 quote chars) Tests
159     // =====
160
161     [Fact]
162     public static void TestTripleDoubleQuotes()
163     {
164         var parser = new Parser();
165         var result = parser.Parse("\"\"\"triple double quotes\"\"\"");
166         Assert.Equal("triple double quotes", GetSingleRefId(result));
167     }
168
169     [Fact]
170     public static void TestTripleDoubleQuotesWithDoubleQuoteInside()
171     {
172         var parser = new Parser();
173         var result = parser.Parse("\"\"\"text with \"\" inside\"\"\"");
174         Assert.Equal("text with \"\" inside", GetSingleRefId(result));
175     }
176
177     [Fact]
178     public static void TestTripleDoubleQuotesWithEscape()
179     {
180         var parser = new Parser();
181         var result = parser.Parse("\"\"\"text with \"\"\"\"\" escaped triple\"\"\"");
182         Assert.Equal("text with \"\"\"\" escaped triple", GetSingleRefId(result));
183     }
184
185     [Fact]
186     public static void TestTripleSingleQuotes()
187     {
188         var parser = new Parser();
189         var result = parser.Parse(''''triple single quotes'''');
190         Assert.Equal("triple single quotes", GetSingleRefId(result));
191     }
192
193     [Fact]

```

```

194 public static void TestTripleBacktickQuotes()
195 {
196     var parser = new Parser();
197     var result = parser.Parse("`triple backtick quotes`");
198     Assert.Equal("triple backtick quotes", GetSingleRefId(result));
199 }
200
201 // =====
202 // Quadruple Quotes (4 quote chars) Tests
203 // =====
204
205 [Fact]
206 public static void TestQuadrupleDoubleQuotes()
207 {
208     var parser = new Parser();
209     var result = parser.Parse("\"\"quadruple double quotes\"");
210     Assert.Equal("quadruple double quotes", GetSingleRefId(result));
211 }
212
213 [Fact]
214 public static void TestQuadrupleSingleQuotes()
215 {
216     var parser = new Parser();
217     var result = parser.Parse('\"quadruple single quotes\"');
218     Assert.Equal("quadruple single quotes", GetSingleRefId(result));
219 }
220
221 [Fact]
222 public static void TestQuadrupleBacktickQuotes()
223 {
224     var parser = new Parser();
225     var result = parser.Parse("`quadruple backtick quotes`");
226     Assert.Equal("quadruple backtick quotes", GetSingleRefId(result));
227 }
228
229 // =====
230 // Quintuple Quotes (5 quote chars) Tests
231 // =====
232
233 [Fact]
234 public static void TestQuintupleDoubleQuotes()
235 {
236     var parser = new Parser();
237     var result = parser.Parse("\"\"\"quintuple double quotes\"");
238     Assert.Equal("quintuple double quotes", GetSingleRefId(result));
239 }
240
241 [Fact]
242 public static void TestQuintupleSingleQuotes()
243 {
244     var parser = new Parser();
245     var result = parser.Parse('\"\"quintuple single quotes\"');
246     Assert.Equal("quintuple single quotes", GetSingleRefId(result));
247 }
248
249 [Fact]
250 public static void TestQuintupleBacktickQuotes()
251 {
252     var parser = new Parser();
253     var result = parser.Parse("`quintuple backtick quotes`");
254     Assert.Equal("quintuple backtick quotes", GetSingleRefId(result));
255 }
256
257 // =====
258 // Complex Scenarios Tests
259 // =====
260
261 [Fact]
262 public static void TestMixedQuotesInLink()
263 {
264     var parser = new Parser();
265     var result = parser.Parse("("double\" 'single' `backtick`)");
266     Assert.Single(result);
267     Assert.NotNull(result[0].Values);
268     Assert.Equal(3, result[0].Values!.Count);
269     Assert.Equal("double", result[0].Values![0].Id);
270     Assert.Equal("single", result[0].Values![1].Id);
271     Assert.Equal("backtick", result[0].Values![2].Id);

```

```

272     }
273
274     [Fact]
275     public static void TestBacktickAsIdInLink()
276     {
277         var parser = new Parser();
278         var result = parser.Parse("`myId`: value1 value2");
279         Assert.Single(result);
280         Assert.Equal("myId", result[0].Id);
281         Assert.NotNull(result[0].Values);
282         Assert.Equal(2, result[0].Values!.Count);
283     }
284
285     [Fact]
286     public static void TestCodeBlockLikeContent()
287     {
288         var parser = new Parser();
289         var result = parser.Parse("```const x = 1;```");
290         Assert.Equal("const x = 1;", GetSingleRefId(result));
291     }
292
293     [Fact]
294     public static void TestNestedQuotesInMarkdown()
295     {
296         var parser = new Parser();
297         var result = parser.Parse("`Use `code` in markdown`");
298         Assert.Equal("Use `code` in markdown", GetSingleRefId(result));
299     }
300
301     [Fact]
302     public static void TestJsonStringWithQuotes()
303     {
304         var parser = new Parser();
305         var result = parser.Parse("{\"key\": \"value\"}");
306         Assert.Equal("{ \"key\": \"value\"}", GetSingleRefId(result));
307     }
308
309     // =====
310     // Edge Cases
311     // =====
312
313     [Fact]
314     public static void TestWhitespacePreservedInQuotes()
315     {
316         var parser = new Parser();
317         var result = parser.Parse(" spaces ");
318         Assert.Equal(" spaces ", GetSingleRefId(result));
319     }
320
321     [Fact]
322     public static void TestMultilineInDoubleDoubleQuotes()
323     {
324         var parser = new Parser();
325         var result = parser.Parse("\"line1\nline2\"");
326         Assert.Single(result);
327         Assert.NotNull(result[0].Values);
328         Assert.Single(result[0].Values!);
329         Assert.Equal("line1\nline2", result[0].Values![0].Id);
330     }
331
332     // =====
333     // Unlimited Quotes (6+ quote chars) Tests
334     // =====
335
336     [Fact]
337     public static void TestUnlimitedQuotes6()
338     {
339         // Test 6-quote strings
340         var parser = new Parser();
341         var result = parser.Parse("\"\"\"\"hello\"\"\"");
342         Assert.Equal("hello", GetSingleRefId(result));
343     }
344
345     [Fact]
346     public static void TestUnlimitedQuotes10()
347     {
348         // Test 10-quote strings
349         var parser = new Parser();

```

```

350         var result = parser.Parse("\"\"\"\"\"\"\"\"\"\"very deeply
↪ quoted\"\"\"\"\"\"\"\"\"\"");
351         Assert.Equal("very deeply quoted", GetSingleRefId(result));
352     }
353
354     [Fact]
355     public static void TestUnlimitedQuotes6WithInnerQuotes()
356     {
357         // Test 6-quote strings with inner 5-quote sequences
358         var parser = new Parser();
359         var result = parser.Parse("\"\"\"\"\"\"\"\"hello with \"\"\"\"\"\" five quotes
↪ inside\"\"\"\"\"\"\"");
360         Assert.Equal("hello with \"\"\"\"\"\" five quotes inside", GetSingleRefId(result));
361     }
362
363     [Fact]
364     public static void TestUnlimitedSingleQuotes7()
365     {
366         // Test 7-quote single quote strings
367         var parser = new Parser();
368         var result = parser.Parse("''''''seven single quotes''''''");
369         Assert.Equal("seven single quotes", GetSingleRefId(result));
370     }
371
372     [Fact]
373     public static void TestUnlimitedBackticks8()
374     {
375         // Test 8-quote backtick strings
376         var parser = new Parser();
377         var result = parser.Parse("`````````eight backticks``````");
378         Assert.Equal("eight backticks", GetSingleRefId(result));
379     }
380 }
381 }

```

1.20 ./Link.Foundation.Links.Notation.Tests/MultilineParserTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MultilineParserTests
7      {
8          [Fact]
9          public static void TwoLinksTest()
10         {
11             var source = @"(first: x y)
12 (second: a b)";
13             var parser = new Parser();
14             var links = parser.Parse(source);
15             var target = links.Format();
16             Assert.Equal(source, target);
17         }
18
19         [Fact]
20         public static void ParseAndStringifyTest()
21         {
22             var source = @"(papa (lovesMama: loves mama))
23 (son lovesMama)
24 (daughter lovesMama)
25 (all (love mama))";
26             var parser = new Parser();
27             var links = parser.Parse(source);
28             var target = links.Format();
29             Assert.Equal(source, target);
30         }
31
32         [Fact]
33         public static void ParseAndStringifyTest2()
34         {
35             var source = @"father (lovesMom: loves mom)
36 son lovesMom
37 daughter lovesMom
38 all (love mom)";
39             var links = new Parser().Parse(source);
40             var target = links.Format(lessParentheses: true);
41             Assert.Equal(source, target);
42         }
43     }

```

```

44     [Fact]
45     public static void ParseAndStringifyWithLessParenthesesTest()
46     {
47         var source = @"lovesMama: loves mama
48 papa lovesMama
49 son lovesMama
50 daughter lovesMama
51 all (love mama)";
52         var parser = new Parser();
53         var links = parser.Parse(source);
54         var target = links.Format(lessParentheses: true);
55         Assert.Equal(source, target);
56     }
57
58     [Fact]
59     public static void DuplicateIdentifiersTest()
60     {
61         var source = @"(a: a b)
62 (a: b c)";
63         var target = @"(a: a b)
64 (a: b c)";
65         var parser = new Parser();
66         var links = parser.Parse(source);
67         var formattedLinks = links.Format();
68         Assert.Equal(target, formattedLinks);
69     }
70
71     [Fact]
72     public static void ComplexStructureTest()
73     {
74         var input = @"(Type: Type Type)
75 Number
76 String
77 Array
78 Value
79 (property: name type)
80 (method: name params return)";
81
82         var result = new Parser().Parse(input);
83         Assert.NotEmpty(result);
84     }
85
86     [Fact]
87     public static void MixedFormatsTest()
88     {
89         // Mix of single-line and multi-line formats
90         var input = @"id1: value1
91 (id2: value2 value3)
92 simple ref
93 (complex:
94     nested1
95     nested2
96 )";
97
98         var result = new Parser().Parse(input);
99         Assert.NotEmpty(result);
100     }
101
102     [Fact]
103     public static void MultilineWithIdTest()
104     {
105         // Test multi-line link with id
106         var input = "(id: value1 value2)";
107         var result = new Parser().Parse(input);
108         Assert.NotEmpty(result);
109     }
110
111     [Fact]
112     public static void MultipleTopLevelElementsTest()
113     {
114         // Test multiple top-level elements
115         var input = "(elem1: val1)\n(elem2: val2)";
116         var result = new Parser().Parse(input);
117         Assert.NotEmpty(result);
118     }
119
120     [Fact]
121     public static void MultilineSimpleLinksTest()
122     {

```

```

123     var input = "(1: 1 1)\n(2: 2 2)";
124     var parser = new Parser();
125     var parsed = parser.Parse(input);
126     Assert.NotEmpty(parsed);
127
128     // Validate regular formatting
129     var output = parsed.Format();
130     Assert.Contains("(1: 1 1)", output);
131     Assert.Contains("(2: 2 2)", output);
132
133     // Validate alternate formatting matches input
134     var outputAlternate = parsed.Format();
135     Assert.Equal(input, outputAlternate);
136 }
137
138 [Fact]
139 public static void IndentedChildrenTest()
140 {
141     var input = "parent\n  child1\n  child2";
142     var parser = new Parser();
143     var parsed = parser.Parse(input);
144
145     // The parsed structure should have parent with children
146     Assert.NotEmpty(parsed);
147 }
148 }
149 }

```

1.21 ./Link.Foundation.Links.Notation.Tests/MultilineQuotedStringTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class MultilineQuotedStringTests
7      {
8          [Fact]
9          public static void TestMultilineDoubleQuotedReference()
10         {
11             var input = @"(
12                 ""long
13                 string literal representing
14                 the reference""
15
16                 'another
17                 long string literal
18                 as another reference'
19             )";
20
21             var parser = new Parser();
22             var result = parser.Parse(input);
23
24             Assert.NotEmpty(result);
25             Assert.Single(result);
26
27             var link = result[0];
28             Assert.Null(link.Id);
29             Assert.NotNull(link.Values);
30             Assert.Equal(2, link.Values.Count);
31
32             // The two references sit on separate lines inside the parentheses, so the
33             // nested context turns each of them into its own link.
34             Assert.Equal(@"long
35 string literal representing
36 the reference", link.Values![0].Values![0].Id);
37
38             Assert.Equal(@"another
39 long string literal
40 as another reference", link.Values![1].Values![0].Id);
41         }
42
43         [Fact]
44         public static void TestSimpleMultilineDoubleQuoted()
45         {
46             var input = @"(""line1
47 line2""");
48             var parser = new Parser();
49             var result = parser.Parse(input);
50
51             Assert.NotEmpty(result);
52             Assert.Single(result);

```



```

52
53     var link = result[0];
54     Assert.Null(link.Id);
55     Assert.NotNull(link.Values);
56     Assert.Single(link.Values);
57     Assert.Equal("line1\nline2", link.Values[0].Id);
58 }
59
60 [Fact]
61 public static void TestSimpleMultilineSingleQuoted()
62 {
63     var input = @"('line1
64 line2')";
65     var parser = new Parser();
66     var result = parser.Parse(input);
67
68     Assert.NotEmpty(result);
69     Assert.Single(result);
70
71     var link = result[0];
72     Assert.Null(link.Id);
73     Assert.NotNull(link.Values);
74     Assert.Single(link.Values);
75     Assert.Equal("line1\nline2", link.Values[0].Id);
76 }
77
78 [Fact]
79 public static void TestMultilineQuotedAsId()
80 {
81     var input = @"("multi
82 line
83 id": value1 value2)";
84     var parser = new Parser();
85     var result = parser.Parse(input);
86
87     Assert.NotEmpty(result);
88     Assert.Single(result);
89
90     var link = result[0];
91     Assert.Equal("multi\nline\nid", link.Id);
92     Assert.NotNull(link.Values);
93     Assert.Equal(2, link.Values.Count);
94 }
95 }
96 }

```

1.22 ./Link.Foundation.Links.Notation.Tests/NestedIndentationTests.cs

```

1  using Xunit;
2
3  namespace Link.Foundation.Links.Notation.Tests
4  {
5      /// <summary>
6      /// Tests for indentation inside parentheses.
7      ///
8      /// See https://github.com/link-foundation/links-notation/issues/282
9      ///
10     /// Indentation is structural at the root, so it must be structural inside parentheses too:
11     /// a parenthesized group opens a nested context that starts fresh at indentation level zero
12     /// and follows exactly the root's rules.
13     /// </summary>
14     public class NestedIndentationTests
15     {
16         private static string FormatSource(string source) => new Parser().Parse(source).Format();
17
18         private static void AssertFormat(string source, string expected) =>
19         ↪ Assert.Equal(expected, FormatSource(source));
20
21         [Fact]
22         public void ParenthesesReproduceRootIndentation()
23         {
24             AssertFormat("a\n b\n c\n d", "(a)\n((a) (b))\n(c)\n((c) (d))");
25
26             // The same lines inside parentheses keep the same structure, nested under
27             // the link the group belongs to.
28             AssertFormat("array (\n a\n b\n c\n d\n)", "(array ((a) ((a) (b)) (c) ((c)
29 ↪ (d))))");
30         }
31     }
32
33     [Fact]

```

```

31 public void ParenthesesKeepRecordBoundaries()
32 {
33     var source = "value (\n id \"1\" \n label \"one\" \n)";
34     Assert.Format(source, "(value ((id 1) (label one)))");
35
36     var links = new Parser().Parse(source);
37     Assert.Single(links);
38
39     var group = links[0].Values![1];
40     Assert.Null(group.Id);
41     Assert.Equal(2, group.Values!.Count);
42     Assert.Equal("id", group.Values[0].Values![0].Id);
43     Assert.Equal("1", group.Values[0].Values![1].Id);
44     Assert.Equal("label", group.Values[1].Values![0].Id);
45     Assert.Equal("one", group.Values[1].Values![1].Id);
46 }
47
48 [Fact]
49 public void ParenthesesKeepSeveralRecordsSeparate()
50 {
51     Assert.Format(
52         "value (\n (id \"1\" label \"one\") \n (id \"2\" label \"two\") \n)",
53         "(value ((id 1 label one) (id 2 label two)))");
54 }
55
56 [Fact]
57 public void ParenthesesNestDeeply()
58 {
59     Assert.Format("outer (\n inner (\n x 1 \n y 2 \n ) \n z 3 \n)", "(outer ((inner
↪ ((x 1) (y 2))) (z 3)))");
60 }
61
62 [Fact]
63 public void SingleLineParenthesesAreUnchanged()
64 {
65     Assert.Format("(a b c)", "(a b c)");
66     Assert.Format("(1: 2 3)", "(1: 2 3)");
67     Assert.Format("(a: b c)", "(a: b c)");
68     Assert.Format("((a b))", "((a b))");
69     Assert.Format("(a)", "(a)");
70     Assert.Format("()", "()");
71 }
72
73 [Fact]
74 public void ParenthesesWithIndentedIdSyntax()
75 {
76     Assert.Format("( \n a: \n b \n c \n)", "(a: b c)");
77 }
78
79 [Fact]
80 public void EmployeeRecordsKeepTheirFields()
81 {
82     var source = "empInfo \n employees: \n ( \n name (James Kirk) \n age 40 \n
↪ ) \n"
83         + " ( \n name (Jean-Luc Picard) \n age 45 \n )";
84     var expected = "(empInfo \n (empInfo) (employees: ((name (James Kirk)) (age 40)))"
85         + " ((name (Jean-Luc Picard)) (age 45)))";
86     Assert.Format(source, expected);
87 }
88 }
89 }

```

1.23 ./Link.Foundation.Links.Notation.Tests/NestedParserTests.cs

```

1 using System;
2 using System.Linq;
3 using Xunit;
4
5 namespace Link.Foundation.Links.Notation.Tests
6 {
7     public static class NestedParserTests
8     {
9         [Fact]
10         public static void SignificantWhitespaceTest()
11         {
12             var source = @"
13 users
14     user1
15         id
16         43

```

```

17         name
18             first
19                 John
20             last
21                 Williams
22         location
23             New York
24         age
25             23
26     user2
27         id
28             56
29         name
30             first
31                 Igor
32             middle
33                 Petrovich
34             last
35                 Ivanov
36         location
37             Moscow
38         age
39             20";
40     var target = @"(users)
41 ((users) (user1))
42 (((users) (user1)) (id))
43 (((users) (user1)) (id)) (43))
44 (((users) (user1)) (name))
45 (((users) (user1)) (name)) (first))
46 (((users) (user1)) (name)) (first)) (John))
47 (((users) (user1)) (name)) (last))
48 (((users) (user1)) (name)) (last)) (Williams))
49 ((users) (user1)) (location))
50 (((users) (user1)) (location)) (New York))
51 ((users) (user1)) (age))
52 (((users) (user1)) (age)) (23))
53 ((users) (user2))
54 (((users) (user2)) (id))
55 (((users) (user2)) (id)) (56))
56 ((users) (user2)) (name))
57 (((users) (user2)) (name)) (first))
58 (((users) (user2)) (name)) (first)) (Igor))
59 (((users) (user2)) (name)) (middle))
60 (((users) (user2)) (name)) (middle)) (Petrovich))
61 (((users) (user2)) (name)) (last))
62 (((users) (user2)) (name)) (last)) (Ivanov))
63 ((users) (user2)) (location))
64 (((users) (user2)) (location)) (Moscow))
65 ((users) (user2)) (age))
66 (((users) (user2)) (age)) (20))";
67     var parser = new Parser();
68     var links = parser.Parse(source);
69     var formattedLinks = links.Format();
70     Assert.Equal(target, formattedLinks);
71 }
72
73 [Fact]
74 public static void SimpleSignificantWhitespaceTest()
75 {
76     var source = @"a
77 b
78 c";
79     var target = @"(a)
80 (a) (b))
81 (a) (c))";
82     var parser = new Parser();
83     var links = parser.Parse(source);
84     var formattedLinks = links.Format();
85     Assert.Equal(target, formattedLinks);
86 }
87
88 [Fact]
89 public static void TwoSpacesSizedWhitespaceTest()
90 {
91     var source = @"
92 users
93     user1";
94     var target = @"(users)
95 ((users) (user1))";
96     var parser = new Parser();

```

```

97         var links = parser.Parse(source);
98         var formattedLinks = links.Format();
99         Assert.Equal(target, formattedLinks);
100     }
101
102     [Fact]
103     public static void ParseNestedStructureWithIndentationTest()
104     {
105         var source = @"parent
106 child1
107 child2";
108         var parser = new Parser();
109         var links = parser.Parse(source);
110         Assert.NotNull(links);
111         // Should create 3 links: (parent), (parent child1), (parent child2)
112         Assert.Equal(3, links.Count);
113     }
114
115     [Fact]
116     public static void IndentationConsistencyTest()
117     {
118         // Test that indentation must be consistent
119         var input = @"parent
120 child1
121 child2"; // Inconsistent indentation
122         var result = new Parser().Parse(input);
123         // This should parse but child2 won't be a child of parent due to different
124         ↪ indentation
125         Assert.NotEmpty(result);
126     }
127
128     [Fact]
129     public static void IndentationBasedChildrenTest()
130     {
131         var input = @"parent
132 child1
133 child2
134   grandchild";
135         var parser = new Parser();
136         var result = parser.Parse(input);
137         Assert.NotNull(result);
138         // Expected flattened links:
139         // (parent), (parent child1), (parent child2), ((parent child2) grandchild)
140         Assert.Equal(4, result.Count);
141     }
142
143     [Fact]
144     public static void ComplexIndentationTest()
145     {
146         var input = @"root
147 level1a
148   level2a
149   level2b
150 level1b
151   level2c";
152         var parser = new Parser();
153         var result = parser.Parse(input);
154         Assert.NotNull(result);
155         // Expected flattened links:
156         // (root), (root level1a), ((root level1a) level2a), ((root level1a) level2b),
157         // (root level1b), ((root level1b) level2c)
158         Assert.Equal(6, result.Count);
159     }
160
161     [Fact]
162     public static void NestedLinksTest()
163     {
164         var input = "(1: (2: (3: 3)))";
165         var parser = new Parser();
166         var parsed = parser.Parse(input);
167         Assert.NotEmpty(parsed);
168
169         // Validate regular formatting
170         var output = parsed.Format();
171         Assert.NotEmpty(output);
172
173         // Validate that the structure is properly nested
174         Assert.Single(parsed);
175     }

```

```

176 [Fact]
177 public static void IndentationParserTest()
178 {
179     var input = "parent\n  child1\n  child2";
180     var parser = new Parser();
181     var result = parser.Parse(input);
182     Assert.NotEmpty(result);
183     // Should have parent link
184     var parentLink = result.FirstOrDefault(l => l.Id == "parent");
185 }
186
187 [Fact]
188 public static void NestedIndentationParserTest()
189 {
190     var input = "parent\n  child\n    grandchild";
191     var parser = new Parser();
192     var result = parser.Parse(input);
193     Assert.NotEmpty(result);
194     // Should create nested structure with proper hierarchy
195     Assert.True(result.Count >= 1);
196 }
197
198 [Fact]
199 public static void ThreeLevelNestingRoundtripTest()
200 {
201     var input = "(1: (2: (3: 3)))";
202     var parser = new Parser();
203     var parsed = parser.Parse(input);
204
205     // Validate round-trip
206     var output = parsed.Format();
207     Assert.Equal(input, output);
208 }
209
210 [Fact]
211 public static void DeepNestedStructureRoundtripTest()
212 {
213     var input = "(a: (b: (c: (d: d))))";
214     var parser = new Parser();
215     var parsed = parser.Parse(input);
216
217     // Validate round-trip
218     var output = parsed.Format();
219     Assert.Equal(input, output);
220 }
221
222 [Fact]
223 public static void MultipleNestedLinksRoundtripTest()
224 {
225     var input = "(parent: (child1: value1) (child2: value2))";
226     var parser = new Parser();
227     var parsed = parser.Parse(input);
228
229     // Validate round-trip
230     var output = parsed.Format();
231     Assert.Equal(input, output);
232 }
233 }
234 }

```

1.24 ./Link.Foundation.Links.Notation.Tests/NestedSelfReferenceTests.cs

```

1 using System;
2 using Xunit;
3
4 namespace Link.Foundation.Links.Notation.Tests
5 {
6     public static class NestedSelfReferenceTests
7     {
8         [Fact]
9         public static void NestedSelfReferencedObjectInPairValueTest()
10        {
11            // Test case from PARSER BUG.md
12            // This should parse a dict with two pairs, where the second pair's value
13            // is itself a self-referenced dict definition (obj 1: dict ...)
14            var notation = "(obj 0: dict ((str bmFtZQ==) (str ZG1jdDE=)) ((str b3RoZXI=) (obj 1:
↵ dict ((str bmFtZQ==) (str ZG1jdDI=)) ((str b3RoZXI=) obj 0))))";
15
16            var parser = new Parser();
17            var links = parser.Parse(notation);

```

```

18 // Should parse exactly one top-level link
19 Assert.Single(links);
20
21 var link = links[0];
22
23 // Top-level link should have ID "obj 0"
24 Assert.Equal("obj 0", link.Id);
25
26 // Should have: type marker + 2 pairs = 3 values
27 Assert.Equal(3, link.Values!.Count);
28
29 // First value is the type marker "dict"
30 Assert.Equal("dict", link.Values[0].Id);
31
32 // Second and third values are the two pairs
33 var pair1 = link.Values[1];
34 var pair2 = link.Values[2];
35
36 // Pair 1: ((str bmFtZQ==) (str ZGljdDE=))
37 // This is a parenthesized expression containing two sub-expressions
38 Assert.Null(pair1.Id);
39 Assert.Equal(2, pair1.Values!.Count);
40
41 // First element of pair1: (str bmFtZQ==)
42 Assert.Null(pair1.Values[0].Id);
43 Assert.Equal(2, pair1.Values![0].Values!.Count);
44 Assert.Equal("str", pair1.Values![0].Values![0].Id);
45 Assert.Equal("bmFtZQ==", pair1.Values![0].Values![1].Id);
46
47 // Second element of pair1: (str ZGljdDE=)
48 Assert.Null(pair1.Values[1].Id);
49 Assert.Equal(2, pair1.Values![1].Values!.Count);
50 Assert.Equal("str", pair1.Values![1].Values![0].Id);
51 Assert.Equal("ZGljdDE=", pair1.Values![1].Values![1].Id);
52
53 // Pair 2: ((str b3RoZXI=) (obj 1: dict ...))
54 // This is a parenthesized expression containing two sub-expressions
55 Assert.Null(pair2.Id);
56 Assert.Equal(2, pair2.Values!.Count);
57
58 // First element of pair2: (str b3RoZXI=)
59 Assert.Null(pair2.Values[0].Id);
60 Assert.Equal(2, pair2.Values![0].Values!.Count);
61 Assert.Equal("str", pair2.Values![0].Values![0].Id);
62 Assert.Equal("b3RoZXI=", pair2.Values![0].Values![1].Id);
63
64 // Second element of pair2: (obj 1: dict ((str bmFtZQ==) (str ZGljdDI=)) ((str
65 ↪ b3RoZXI=) obj 0))
66 // THIS IS THE KEY TEST - obj 1 should have its ID preserved
67 var obj1 = pair2.Values[1];
68 Assert.Equal("obj 1", obj1.Id);
69 Assert.NotNull(obj1.Values);
70 Assert.Equal(3, obj1.Values.Count); // type marker + 2 pairs
71
72 // obj 1's type marker
73 Assert.Equal("dict", obj1.Values[0].Id);
74
75 // obj 1's first pair: ((str bmFtZQ==) (str ZGljdDI=))
76 var obj1Pair1 = obj1.Values[1];
77 Assert.Equal(2, obj1Pair1.Values!.Count);
78 Assert.Null(obj1Pair1.Values[0].Id);
79 Assert.Equal(2, obj1Pair1.Values![0].Values!.Count);
80 Assert.Equal("str", obj1Pair1.Values![0].Values![0].Id);
81 Assert.Equal("bmFtZQ==", obj1Pair1.Values![0].Values![1].Id);
82 Assert.Null(obj1Pair1.Values[1].Id);
83 Assert.Equal(2, obj1Pair1.Values![1].Values!.Count);
84 Assert.Equal("str", obj1Pair1.Values![1].Values![0].Id);
85 Assert.Equal("ZGljdDI=", obj1Pair1.Values![1].Values![1].Id);
86
87 // obj 1's second pair: ((str b3RoZXI=) obj 0) - reference back to obj 0
88 var obj1Pair2 = obj1.Values[2];
89 Assert.Equal(2, obj1Pair2.Values!.Count);
90 Assert.Null(obj1Pair2.Values[0].Id);
91 Assert.Equal(2, obj1Pair2.Values![0].Values!.Count);
92 Assert.Equal("str", obj1Pair2.Values![0].Values![0].Id);
93 Assert.Equal("b3RoZXI=", obj1Pair2.Values![0].Values![1].Id);
94 Assert.Equal("obj 0", obj1Pair2.Values[1].Id);
95 Assert.Null(obj1Pair2.Values[1].Values); // Just a reference, no nested values

```

```

96     }
97
98     [Fact]
99     public static void SelfReferenceAsDirectChildWorksCorrectlyTest()
100    {
101        // This pattern should work (and did work before)
102        var notation = "(obj 0: list (int 1) (int 2) (obj 1: list (int 3) (int 4) obj 0))";
103
104        var parser = new Parser();
105        var links = parser.Parse(notation);
106
107        Assert.Single(links);
108        Assert.Equal("obj 0", links[0].Id);
109        Assert.Equal(4, links[0].Values!.Count); // list + 1 + 2 + obj 1
110
111        // The fourth value should be obj 1 with a self-reference
112        var obj1 = links[0].Values![3];
113        Assert.Equal("obj 1", obj1.Id);
114        Assert.Equal(4, obj1.Values!.Count); // list + 3 + 4 + obj 0
115        Assert.Equal("obj 0", obj1.Values[3].Id); // Reference to obj 0
116    }
117 }
118 }

```

1.25 ./Link.Foundation.Links.Notation.Tests/SingleLineParserTests.cs

```

1  using System;
2  using Xunit;
3
4  namespace Link.Foundation.Links.Notation.Tests
5  {
6      public static class SingleLineParserTests
7      {
8          [Fact]
9          public static void SingleLinkTest()
10         {
11             var source = @"(address: source target)";
12             var parser = new Parser();
13             var links = parser.Parse(source);
14             var target = links.Format();
15             Assert.Equal(source, target);
16         }
17
18         [Fact]
19         public static void TripletSingleLinkTest()
20         {
21             var source = @"(papa has car)";
22             var parser = new Parser();
23             var links = parser.Parse(source);
24             var target = links.Format();
25             Assert.Equal(source, target);
26         }
27
28         [Fact]
29         public static void BugTest1()
30         {
31             var source = @"(ignore conan-center-index repository)";
32             var links = new Parser().Parse(source);
33             var target = links.Format();
34             Assert.Equal(source, target);
35         }
36
37         [Fact]
38         public static void QuotedReferencesTest()
39         {
40             var source = @"(a: 'b' ""c"")";
41             var target = @"(a: b c)";
42             var parser = new Parser();
43             var links = parser.Parse(source);
44             var formattedLinks = links.Format();
45             Assert.Equal(target, formattedLinks);
46         }
47
48         [Fact]
49         public static void QuotedReferencesWithSpacesTest()
50         {
51             var source = @"('a a': 'b b' ""c c"")";
52             var target = @"('a a': 'b b' 'c c')";
53             var parser = new Parser();
54             var links = parser.Parse(source);

```

```

55     var formattedLinks = links.Format();
56     Assert.Equal(target, formattedLinks);
57 }
58
59 [Fact]
60 public static void ParseSimpleReferenceTest()
61 {
62     var source = "test";
63     var parser = new Parser();
64     var links = parser.Parse(source);
65     Assert.NotNull(links);
66     Assert.Single(links);
67     // Simple reference creates a singlet link with null Id and one value
68     Assert.Null(links[0].Id);
69     Assert.NotNull(links[0].Values);
70     Assert.Single(links[0].Values!);
71     Assert.Equal("test", links[0].Values?[0].Id);
72     Assert.Null(links[0].Values?[0].Values);
73 }
74
75 [Fact]
76 public static void ParseReferenceWithColonAndValuesTest()
77 {
78     var source = "parent: child1 child2";
79     var parser = new Parser();
80     var links = parser.Parse(source);
81     Assert.NotNull(links);
82     Assert.Single(links);
83     Assert.Equal("parent", links[0].Id);
84     Assert.NotNull(links[0].Values);
85     Assert.Equal(2, links[0].Values!.Count);
86     Assert.Equal("child1", links[0].Values![0].Id);
87     Assert.Equal("child2", links[0].Values![1].Id);
88 }
89
90 [Fact]
91 public static void ParseMultilineLinkTest()
92 {
93     var source = "(parent: child1 child2)";
94     var parser = new Parser();
95     var links = parser.Parse(source);
96     Assert.NotNull(links);
97     Assert.Single(links);
98     Assert.Equal("parent", links[0].Id);
99     Assert.NotNull(links[0].Values);
100    Assert.Equal(2, links[0].Values!.Count);
101 }
102
103 [Fact]
104 public static void ParseValuesOnlyTest()
105 {
106     var source = ": value1 value2";
107     var parser = new Parser();
108     // Standalone ':' is now forbidden and should throw an exception
109     Assert.Throws<FormatException>(() => parser.Parse(source));
110 }
111
112 [Fact]
113 public static void SingletLinkTest()
114 {
115     // Test singlet link
116     var input = "(singlet)";
117     var result = new Parser().Parse(input);
118     Assert.Single(result);
119     Assert.Null(result[0].Id);
120     Assert.NotNull(result[0].Values);
121     Assert.Single(result[0].Values!);
122     Assert.Equal("singlet", result[0].Values?[0].Id);
123     Assert.Null(result[0].Values?[0].Values);
124 }
125
126 [Fact]
127 public static void ValueLinkTest()
128 {
129     // Test value link
130     var input = "(value1 value2 value3)";
131     var result = new Parser().Parse(input);
132     Assert.NotEmpty(result);

```



```

133     }
134
135     [Fact]
136     public static void QuotedReferencesWithSpecialCharsTest()
137     {
138         // Test quoted references
139         var input = @"("id with spaces": "value with spaces")";
140         var result = new Parser().Parse(input);
141         Assert.NotEmpty(result);
142     }
143
144     [Fact]
145     public static void SingleQuotedReferencesTest()
146     {
147         // Test single-quoted references
148         var input = "('id': 'value')";
149         var result = new Parser().Parse(input);
150         Assert.NotEmpty(result);
151     }
152
153     [Fact]
154     public static void ParseQuotedReferencesValuesOnlyTest()
155     {
156         var source = @"\has space\" 'has:colon';
157         var parser = new Parser();
158         var links = parser.Parse(source);
159         Assert.NotNull(links);
160         Assert.Single(links);
161         Assert.Null(links[0].Id);
162         Assert.NotNull(links[0].Values);
163         Assert.Equal(2, links[0].Values!.Count);
164         Assert.Equal("has space", links[0].Values![0].Id);
165         Assert.Equal("has:colon", links[0].Values![1].Id);
166         var formatted = links.Format();
167         Assert.Equal("('has space' 'has:colon')", formatted);
168     }
169
170     [Fact]
171     public static void NestedLinksTest()
172     {
173         // Test nested links
174         var input = "(outer: (inner: value))";
175         var result = new Parser().Parse(input);
176         Assert.NotEmpty(result);
177     }
178
179     [Fact]
180     public static void HyphenatedIdentifiersTest()
181     {
182         // Test support for hyphenated identifiers like in BugTest1
183         var source = @"(conan-center-index: repository info)";
184         var parser = new Parser();
185         var links = parser.Parse(source);
186         var target = links.Format();
187         Assert.Equal(source, target);
188     }
189
190     [Fact]
191     public static void MultipleWordsInQuotesTest()
192     {
193         var source = @"("New York": city state)";
194         var parser = new Parser();
195         var links = parser.Parse(source);
196         var target = links.Format();
197         // Should preserve quotes for multi-word references
198         Assert.Contains("New York", target);
199     }
200
201     [Fact]
202     public static void SpecialCharactersInQuotesTest()
203     {
204         var input = @"("key:with:colons": "value(with)parentheses")";
205         var result = new Parser().Parse(input);
206         Assert.NotEmpty(result);
207
208         input = @"('key with spaces': 'value: with special chars')";
209         result = new Parser().Parse(input);
210         Assert.NotEmpty(result);

```

```

211     }
212
213     [Fact]
214     public static void DeeplyNestedTest()
215     {
216         var input = "(a: (b: (c: (d: (e: value))))";
217         var result = new Parser().Parse(input);
218         Assert.NotEmpty(result);
219     }
220
221     [Fact]
222     public static void SingleLineWithIdTest()
223     {
224         // Test single-line link with id
225         var input = "id: value1 value2";
226         var result = new Parser().Parse(input);
227         Assert.NotEmpty(result);
228     }
229
230     [Fact]
231     public static void SingleLineWithoutIdTest()
232     {
233         // Test link without id (single-line) - now forbidden
234         var input = ": value1 value2";
235         Assert.Throws<FormatException>(() => new Parser().Parse(input));
236     }
237
238     [Fact]
239     public static void MultilineWithoutIdTest()
240     {
241         // Test link without id (multi-line) - now forbidden
242         var input = "(: value1 value2)";
243         Assert.Throws<FormatException>(() => new Parser().Parse(input));
244     }
245
246     [Fact]
247     public static void SimpleRefTest()
248     {
249         var input = "simple ref";
250         var result = new Parser().Parse(input);
251         Assert.NotEmpty(result);
252     }
253
254     [Fact]
255     public static void MultiLineLinkWithIdTest()
256     {
257         var input = "(id: value1 value2)";
258         var result = new Parser().Parse(input);
259         Assert.NotEmpty(result);
260     }
261
262     [Fact]
263     public static void LinkWithoutIdMultiLineTest()
264     {
265         var input = "(: value1 value2)";
266         Assert.Throws<FormatException>(() => new Parser().Parse(input));
267     }
268
269     [Fact]
270     public static void SimpleReferenceParserTest()
271     {
272         var input = "hello";
273         var result = new Parser().Parse(input);
274         Assert.Single(result);
275         Assert.Null(result[0].Id);
276         Assert.NotNull(result[0].Values);
277         Assert.Single(result[0].Values!);
278         Assert.Equal("hello", result[0].Values?[0].Id);
279     }
280
281     [Fact]
282     public static void QuotedReferenceParserTest()
283     {
284         var input = "\"hello world\"";
285         var result = new Parser().Parse(input);
286         Assert.Single(result);
287         Assert.Null(result[0].Id);
288         Assert.NotNull(result[0].Values);
289         Assert.Single(result[0].Values!);

```

```

290     Assert.Equal("hello world", result[0].Values?[0].Id);
291 }
292
293 [Fact]
294 public static void SingletLinkParserTest()
295 {
296     var input = "(singlet)";
297     var result = new Parser().Parse(input);
298     Assert.Single(result);
299     Assert.Null(result[0].Id);
300     Assert.NotNull(result[0].Values);
301     Assert.Single(result[0].Values!);
302     Assert.Equal("singlet", result[0].Values?[0].Id);
303     Assert.Null(result[0].Values?[0].Values);
304 }
305
306 [Fact]
307 public static void ValueLinkParserTest()
308 {
309     var input = "(a b c)";
310     var result = new Parser().Parse(input);
311     Assert.Single(result);
312     Assert.Null(result[0].Id);
313     Assert.Equal(3, result[0].Values?.Count);
314 }
315
316 [Fact]
317 public static void ParseQuotedReferencesTest()
318 {
319     var input = "(\"quoted id\": \"value with spaces\")";
320     var parser = new Parser();
321     var result = parser.Parse(input);
322
323     Assert.NotEmpty(result);
324     var formatted = result.Format();
325     Assert.Contains("quoted id", formatted);
326     Assert.Contains("value with spaces", formatted);
327 }
328
329 [Fact]
330 public static void LinkWithoutIdSingleLineTest()
331 {
332     // Test that standalone ':' is forbidden and should throw
333     var input = ": value1 value2";
334     var parser = new Parser();
335
336     // C# parser forbids this syntax (like JS/Rust)
337     Assert.Throws<FormatException>(() => parser.Parse(input));
338 }
339
340 [Fact]
341 public static void SingleLineLinkWithIdTest()
342 {
343     var input = "id: value1 value2";
344     var parser = new Parser();
345     var result = parser.Parse(input);
346
347     Assert.NotEmpty(result);
348     Assert.NotNull(result[0].Id);
349     Assert.Equal("id", result[0].Id);
350 }
351
352 [Fact]
353 public static void LinkWithIdTest()
354 {
355     var input = "(id: a b c)";
356     var parser = new Parser();
357     var result = parser.Parse(input);
358
359     Assert.Single(result);
360     Assert.Equal("id", result[0].Id);
361     Assert.Equal(3, result[0].Values?.Count);
362 }
363
364 [Fact]
365 public static void QuotedReferenceTest()
366 {
367     var input = "\"quoted value\"";
368     var parser = new Parser();

```

```

369         var result = parser.Parse(input);
370
371         Assert.NotEmpty(result);
372         Assert.NotNull(result[0].Values);
373         Assert.Single(result[0].Values!);
374         Assert.Equal("quoted value", result[0].Values![0].Id);
375     }
376
377     [Fact]
378     public static void SimpleReferenceTest()
379     {
380         var input = "simplereference";
381         var parser = new Parser();
382         var result = parser.Parse(input);
383
384         Assert.NotEmpty(result);
385     }
386
387     [Fact]
388     public static void SingleLineLinkTest()
389     {
390         var input = "myid: value1 value2";
391         var parser = new Parser();
392         var result = parser.Parse(input);
393
394         Assert.Single(result);
395         Assert.Equal("myid", result[0].Id);
396         Assert.Equal(2, result[0].Values?.Count);
397     }
398
399     [Fact]
400     public static void ValuesOnlyInParenthesesTest()
401     {
402         var input = "(value1 value2)";
403         var parser = new Parser();
404         var result = parser.Parse(input);
405
406         Assert.Single(result);
407         Assert.Null(result[0].Id);
408         Assert.Equal(2, result[0].Values?.Count);
409     }
410
411     [Fact]
412     public static void ParseValuesOnlyStandaloneColonTest()
413     {
414         // Test that standalone ':' is forbidden and should throw
415         var input = ": value1 value2";
416         var parser = new Parser();
417
418         // C# parser forbids this syntax (like JS/Rust)
419         Assert.Throws<FormatException>(() => parser.Parse(input));
420     }
421
422     [Fact]
423     public static void QuotedReferencesWithSpacesInLinkTest()
424     {
425         var input = "(id: \"value with spaces\")";
426         var parser = new Parser();
427         var result = parser.Parse(input);
428
429         Assert.Single(result);
430         Assert.Equal("id", result[0].Id);
431         Assert.Single(result[0].Values!);
432         Assert.Equal("value with spaces", result[0].Values?[0].Id);
433     }
434 }
435 }

```

1.26 ./Link.Foundation.Links.Notation.Tests/TupleTests.cs

```

1  using System.Collections.Generic;
2  using Xunit;
3  using LinkType = Link.Foundation.Links.Notation.Link<string>;
4  using id = Link.Foundation.Links.Notation.id<string>;
5  using = Link.Foundation.Links.Notation. <string>;
6
7  namespace Link.Foundation.Links.Notation.Tests
8  {
9      public class TupleTests
10     {
11         [Fact]

```

```

12     public void TupleToLinkTest()
13     {
14         var source = @"(papa (lovesMama: loves mama))
15 (son lovesMama)
16 (daughter lovesMama)
17 (all (love mama))";
18         var parser = new Parser();
19         var links = parser.Parse(source);
20         var targetFromString = links.Format();
21
22         IList<LinkType> constructedLinks = new List<LinkType>()
23         {
24             ("papa", ( )("lovesMama", "loves", "mama")),
25             ("son", "lovesMama"),
26             ("daughter", "lovesMama"),
27             ("all", ("love", "mama")),
28         };
29         var targetFromTuples = constructedLinks.Format();
30         Assert.Equal(targetFromString, targetFromTuples);
31     }
32
33     [Fact]
34     public void NamedTupleToLinkTest()
35     {
36         var source = @"(papa (lovesMama: loves mama))
37 (son lovesMama)
38 (daughter lovesMama)
39 (all (love mama))";
40         var parser = new Parser();
41         var links = parser.Parse(source);
42         var targetFromString = links.Format();
43
44         IList<LinkType> constructedLinks = new List<LinkType>()
45         {
46             ("papa", ((id)"lovesMama", "loves", "mama")),
47             ("son", "lovesMama"),
48             ("daughter", "lovesMama"),
49             ("all", ("love", "mama")),
50         };
51         var targetFromTuples = constructedLinks.Format();
52         Assert.Equal(targetFromString, targetFromTuples);
53     }
54 }
55 }

```

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